AI ACADEMY · LEVEL 3 · GRADE 8 · WEEK 34 · STUDENT LESSON

Capstone Build

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Verified Study Buddy

THE CONNECTED WEEK

Build the stages. Preserve the facts. Check every added result.

Implement a small deterministic report path without inventing missing measurements or ratios.

Keep one habit: Report contract → E2 claim trace → complete baseline program → five integration cases → independent seed repair.

MAIN LESSON · SECTION 1

Does the build preserve the evidence in the brief?

> Connect one full path and inspect every added claim.

Your learning goals

I can explain the **mechanism** and terminology of capstone grounded assistant.

I can apply the stated procedure to a new case with a traceable result.

I can identify an assumption or limitation and design a revealing follow-up.

Recall the requirement-to-test chain

- Close your notes first. A rough first answer helps us choose the right starting point; it is not a score for your ability.

- Recall Capstone Design: a requirement needs permitted sources, output rules and a linked test. Build a small path that preserves those decisions and record what actually happened, including missing results.

Your end goal

By the end, I can explain capstone build using a traced example and a stated limitation.

▼ Earlier wording, alternatives and picture transcripts

Your learning target

Remember before reading

Without opening the old lesson, explain one key idea from Capstone Design. Give an example and a mistake that the idea helps you avoid.

Without opening the old lesson, explain one key idea from Safety Boundaries. Give an example and a mistake that the idea helps you avoid.

Without opening the old lesson, explain one key idea from Privacy by Design. Give an example and a mistake that the idea helps you avoid.

After trying, check the relevant lesson or ask your teacher. Change the part of your explanation that the **evidence** shows was wrong.

Week 34: Capstone Build

Implementation should preserve **traceability** from requirement to feature.



CREATOR MISSION

Build the smallest complete Study Buddy workflow.

Creator name and date:

> **Write or draw your reasoning and evidence.**

Readiness for the connected explanation

This week’s end goal

MAIN LESSON · SECTION 2

Implement the smallest complete source-to-answer path

> A component should satisfy a named requirement and have a check.

Understand traceable implementation

A capstone build turns the design into inspectable stages. Keep a record of the input, eligible source, parsed facts, answerability, draft, validation and displayed outcome. Start with a deterministic baseline that preserves source facts before adding generated wording. A successful build step does not prove an experiment’s result, and a procedure description does not supply measurements.

Paper simulation remains useful when labelled; actual execution needs its own observed output.

Build the assistant as a sequence of testable stages: validate request, retrieve authorised **evidence**, decide answerability, draft, validate and return. Avoid merging all errors into one model score.

Use supplied fictional evidence

Begin by writing the question the procedure is intended to answer. Identify the information already supplied, the information it will produce and the information still missing.

These are constructed classroom examples; they do not require live accounts, private records or an external service. Do not treat a plausible explanation as a measured outcome.

▼ Earlier wording, alternatives and picture transcripts

PICTURE STORY

Notice → Predict → Explain

Begin with visible **evidence**. Do not explain more than the picture and information support.



MYSTERY

The learners must use capstone build to solve a problem. What information do they have? What decision or calculation do they make? What would count as **evidence** that it worked?

Three **observation**s (what you can point to):

> **Write or draw your reasoning and evidence.**

My **prediction** and the clue supporting it:

> **Write or draw your reasoning and evidence.**

A closer explanation

MAIN LESSON · SECTION 3

State what each field can establish

> Keep procedure facts, measured loss and ratio claims separate.

Complete report contract

Investigation-report contract	Declared behavior
Source E2	Three fictional cups observed for four days; no water-loss measurements or initial volumes
Changed source E3	Same three cups/four days, losses 1, 2, 3 mL; initial volumes still unavailable
Procedure claims	Retain cup count, duration and source ID
Measured outcome	Use only supplied losses; None means unavailable, not zero

Investigation-report contract	Declared behavior
Half-water claim	Needs each cup’s loss and corresponding positive initial volume; losses alone cannot establish the ratio
Universal claim	Neither toy record proves effectiveness for all conditions
Implementation limit	The program validates short structured records and builds a report; it is not a general natural-language fact checker or measurement-authenticity test

▼ Separate room-approval review contract

A supported response requires current matching approval and register records. Any retrieval or validation failure gives needs_review with null claims. The classroom implementation performs no booking or external action.

Respect missing-input limits

Before applying the procedure, write its units, denominator or decision boundary where relevant. A partner should be able to repeat each step using the same **input**s.

If a required **input** is absent or violates the stated contract, record that condition explicitly instead of quietly replacing it with a convenient value.

Explain the deciding check

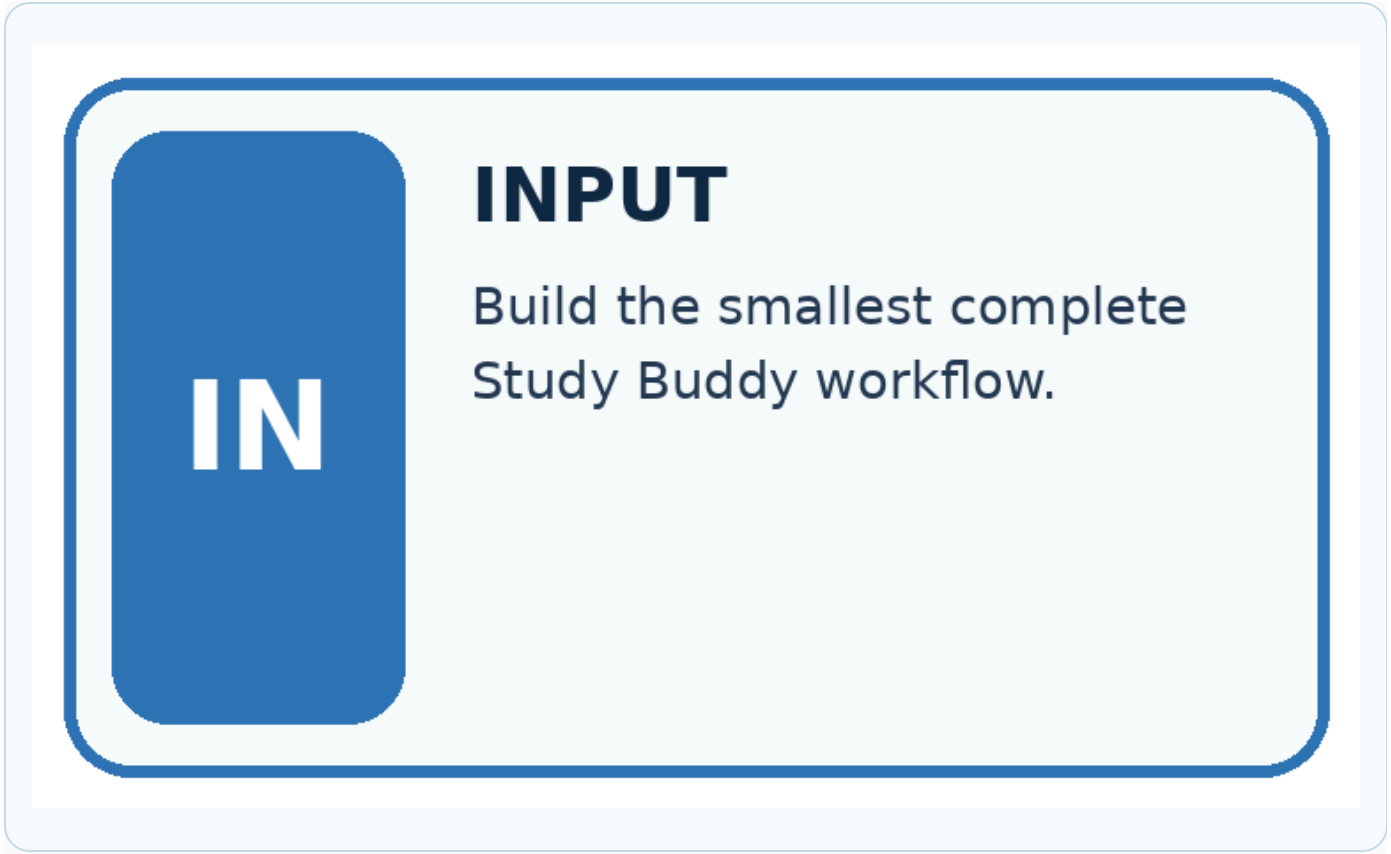
- Point to the **input**, the deciding step and the result in this week’s visual. Explain the same step in ordinary words. A picture helps when you can say what each part represents.

▼ Earlier wording, alternatives and picture transcripts

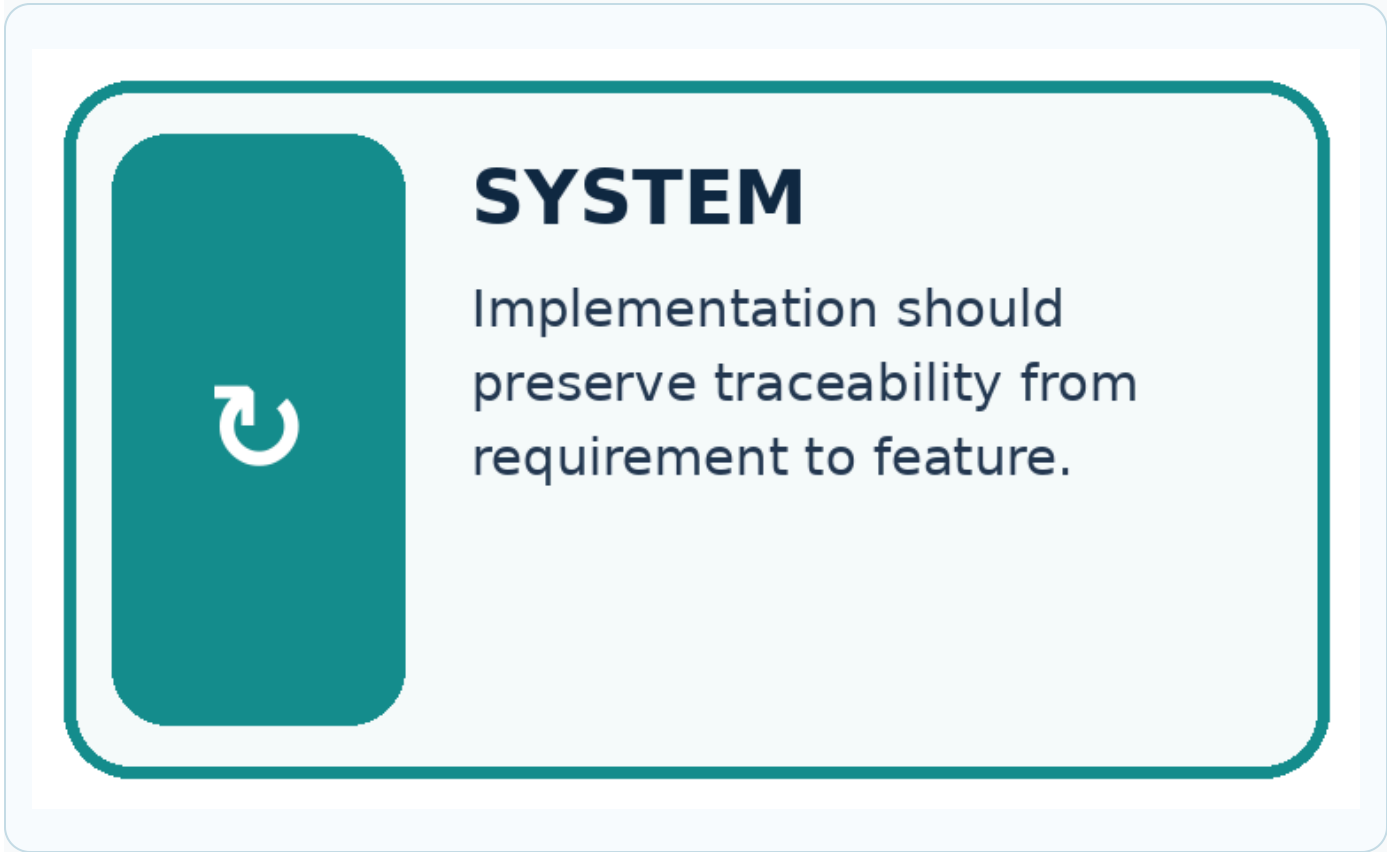
VISUAL **MECHANISM**

Input → **Process** → **Output**

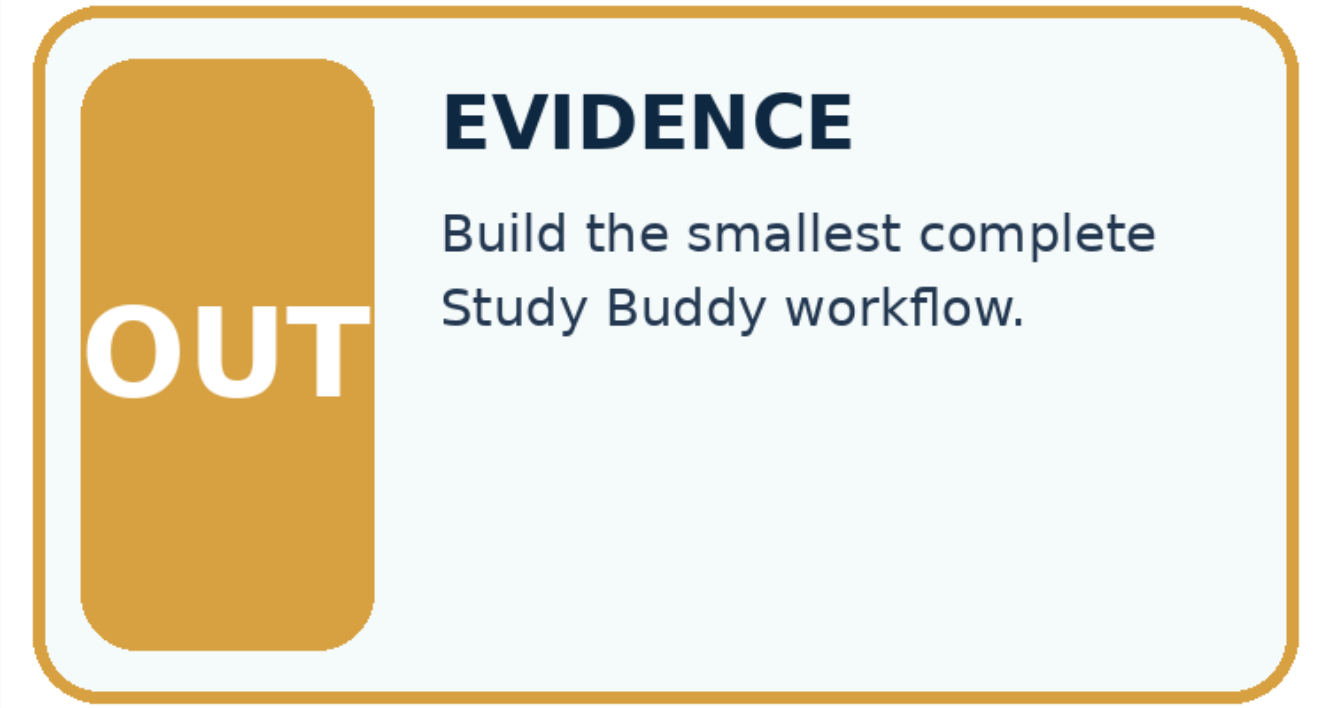
Trace the information. At each stage, ask what changed and how you could test it.


IN
INPUT
Build the smallest complete Study Buddy workflow.

↓


SYSTEM
Implementation should preserve traceability from requirement to feature.

↓



Which stage could fail even when the other two are correct? Explain:

Write or draw your reasoning and evidence.

A closer explanation

Capstone grounded assistant — evidence trace

INPUT AND SCOPE	CHECK OR CALCULATION	CONCLUSION AND LIMIT
Request room B; retrieved approval names B but register names C. Draft gives capacity 10 from C.	A supported response requires current matching approval and register records. Any retrieval or validation failure gives needs_review with null claims. The classroom implementation performs no booking or external action.	Same-room support fails before or during semantic validation. Return needs_review with null claims; the draft capacity is unsupported for B.

The figure connects this chapter's worked input, declared procedure and result. Read the complete worked reasoning for qualifiers and intermediate steps; the abbreviated labels do not replace that reasoning.

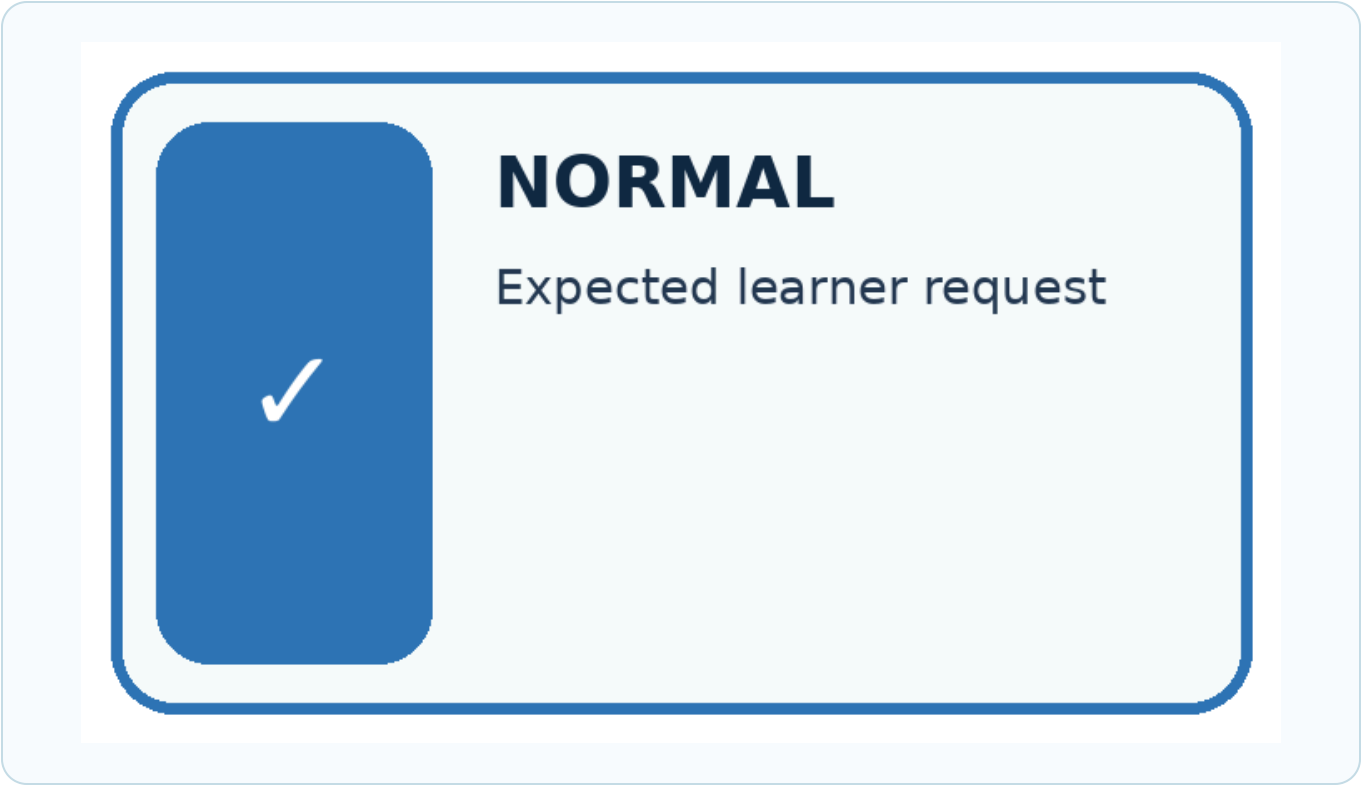
Connect the picture and the explanation

MAIN LESSON · SECTION 4

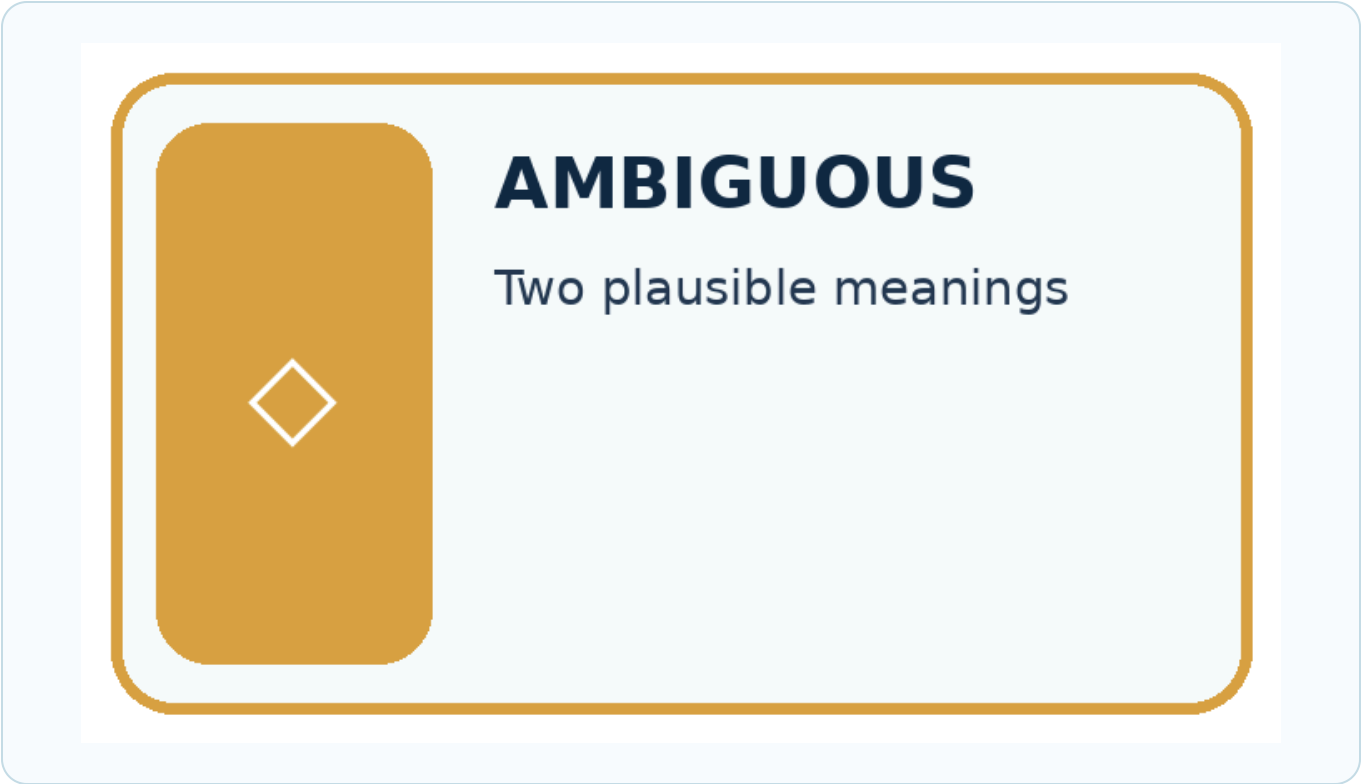
Compare a recorded procedure with an invented outcome

> An observation period is not a measured success result.

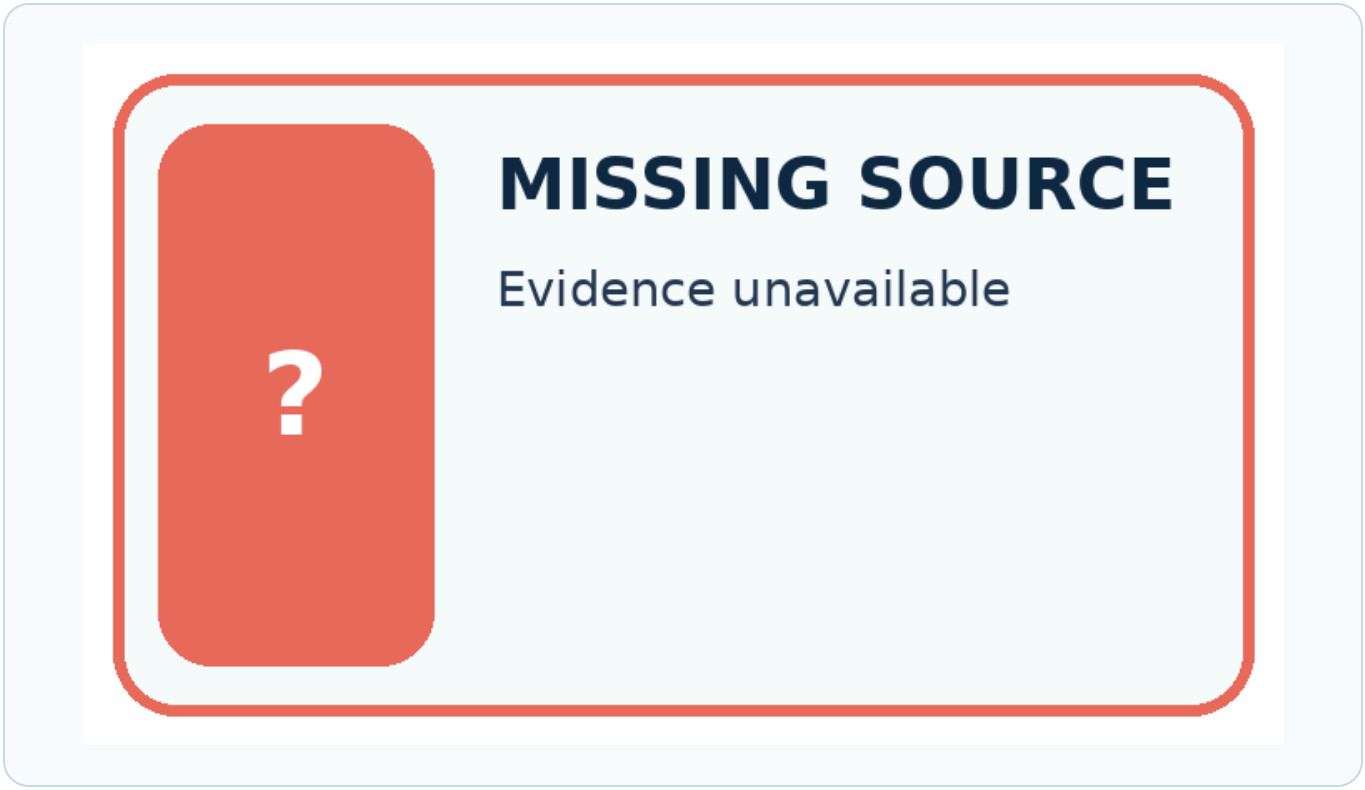
Four pictures with a purpose



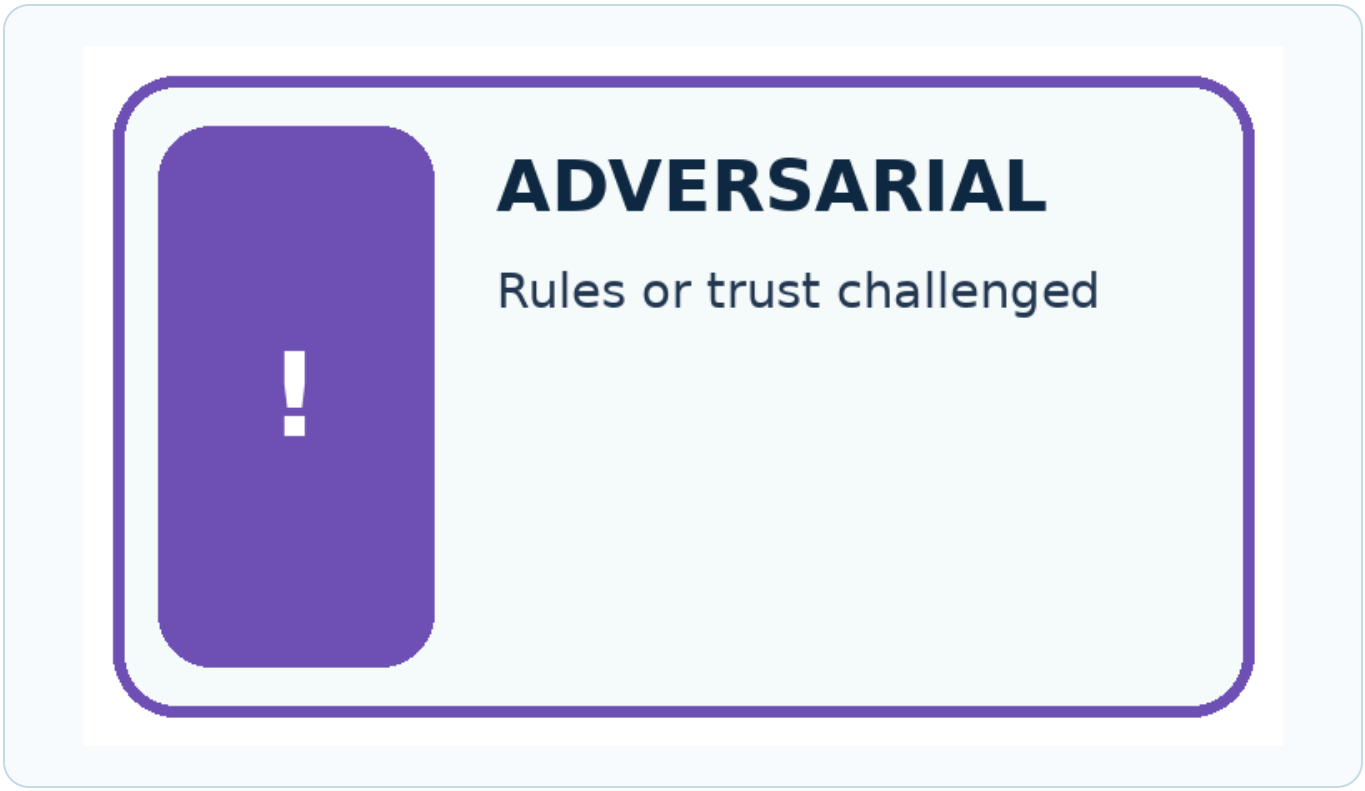
Normal case: predict the result and deciding evidence.



Boundary case: identify the exact limit.



Missing information: name what must be clarified.



Transfer case: explain which procedure still applies.

Choose a revealing case

- Which case is most likely to reveal a hidden assumption? Why?

> **Write or draw your reasoning and evidence.**

Predict the supplied changed case

- Change: E3 supplies losses of 1, 2 and 3 mL.

▼ Check after predicting

Prediction: Report those individual losses, not “half” unless starting volumes are also available.

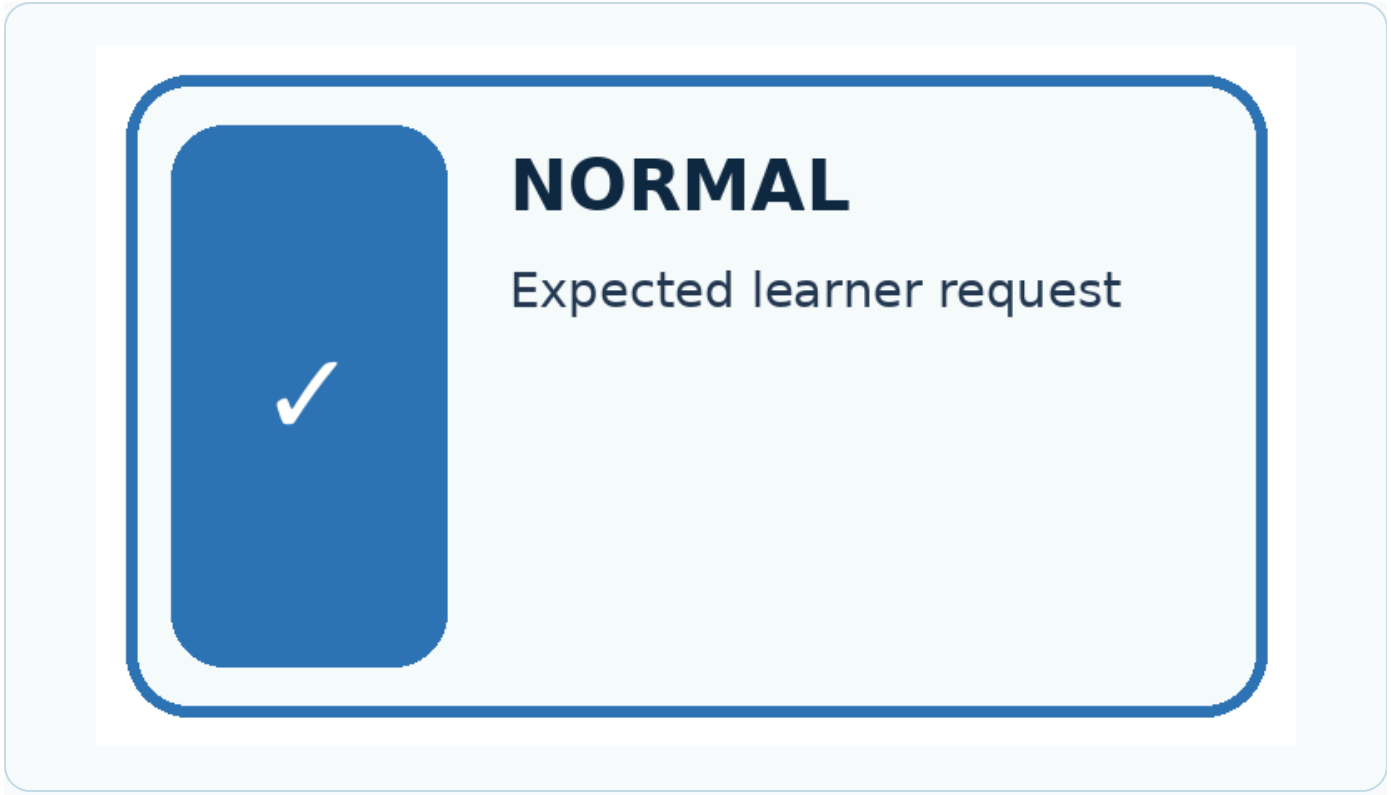
Reason: Absolute losses alone do not establish each cup’s fractional loss.

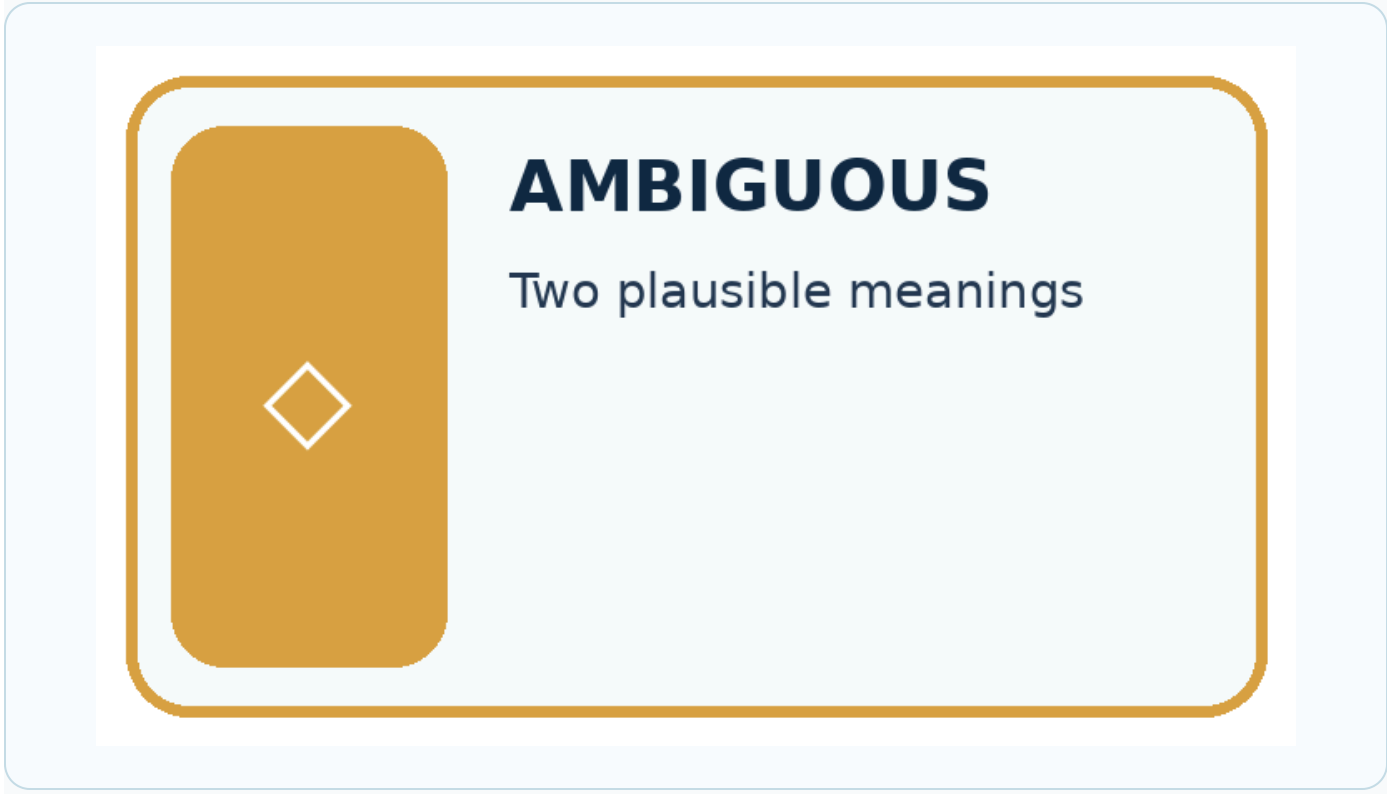
▼ Earlier wording, alternatives and picture transcripts

CASE GALLERY

Four Tests, Four Kinds of **Evidence**

A system is not understood until it survives more than one comfortable example.





NORMAL CASE: predict and name **evidence**. BOUNDARY CASE: predict and name **evidence**.





MISSING INFORMATION: predict and name **evidence**. TRANSFER CASE: predict and name **evidence**.

Change one thing in the pictured case

MAIN LESSON · SECTION 6

Build the E2 report without inventing water loss

> Validate the supplied structure before formatting a conclusion.

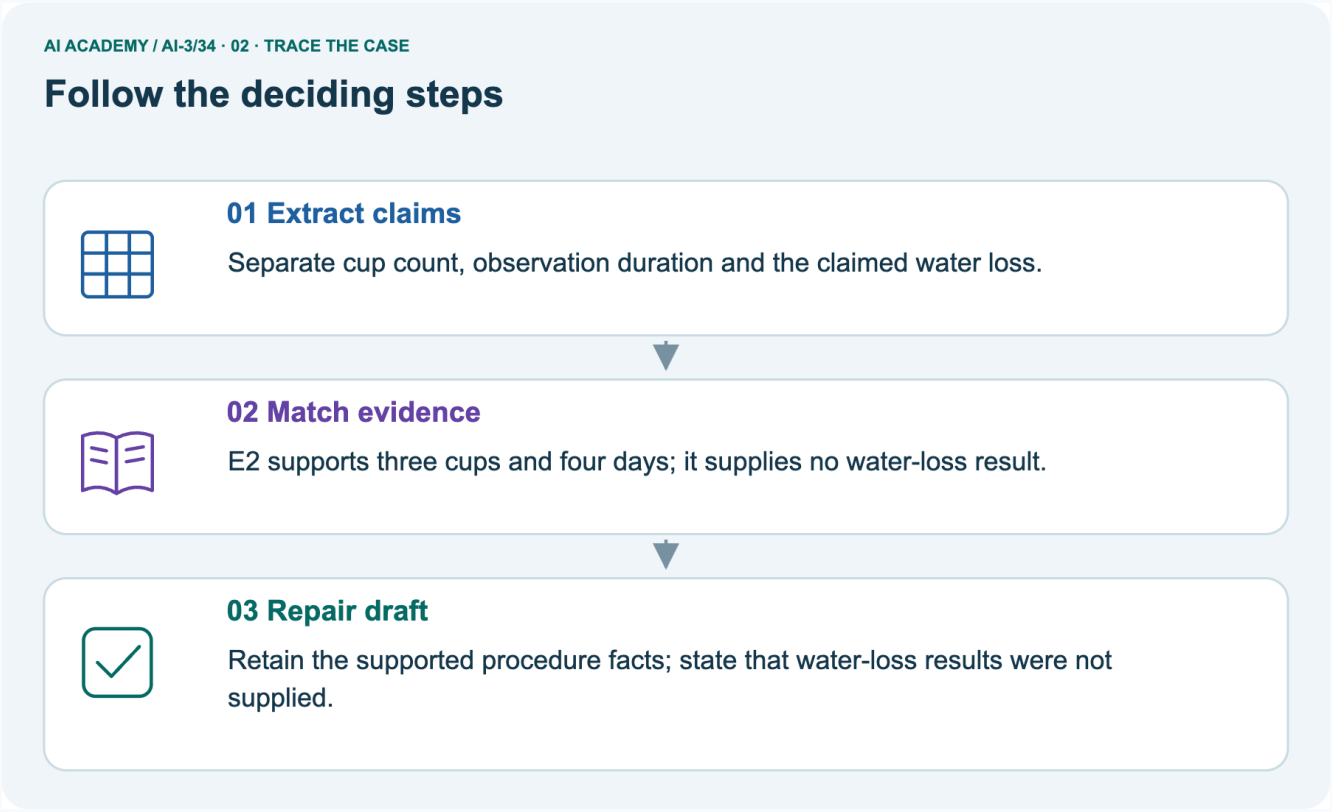
Read the full E2 record and draft

Build a claim checker for a draft summary of a fictional class investigation.

Approved record E2: three cups were observed for four days. No measured water loss is provided. Draft says “three cups were observed for four days and all lost half their water”.

All **input**s and outcomes in this worked example are invented classroom data; no real model run is claimed.

Trace claim extraction, support and repair



Follow the labelled picture with the matching explanation. Open the vector diagram

Read the picture one step at a time

STEP	WHAT HAPPENS AND WHY
1. Extract claims	Separate cup count, observation duration and the claimed water loss.
2. Match evidence	E2 supports three cups and four days; it supplies no water-loss result.
3. Repair draft	Retain the supported procedure facts; state that water-loss results were not supplied.

Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

Read the picture in order. Retain the supported procedure facts; state that water-loss results were not supplied. The arrows show which recorded **input** or intermediate result each decision uses.

Explain the deciding step aloud before checking the changed case; pointing to the final answer alone does not show how it follows.

Run or trace the complete deterministic baseline

In the structured toy program, cups/days are positive integers; optional measurements are lists of nonnegative whole mL values, one per cup. Initial volumes, when supplied, must be positive whole mL and at least the corresponding losses. These deliberately narrow assumptions are not a general measurement schema. E2 produces procedure facts with unavailable losses/half status.

E3 adds 1, 2, 3 mL losses but half remains unknown without initial volumes. No program output proves the source record was reliably measured.

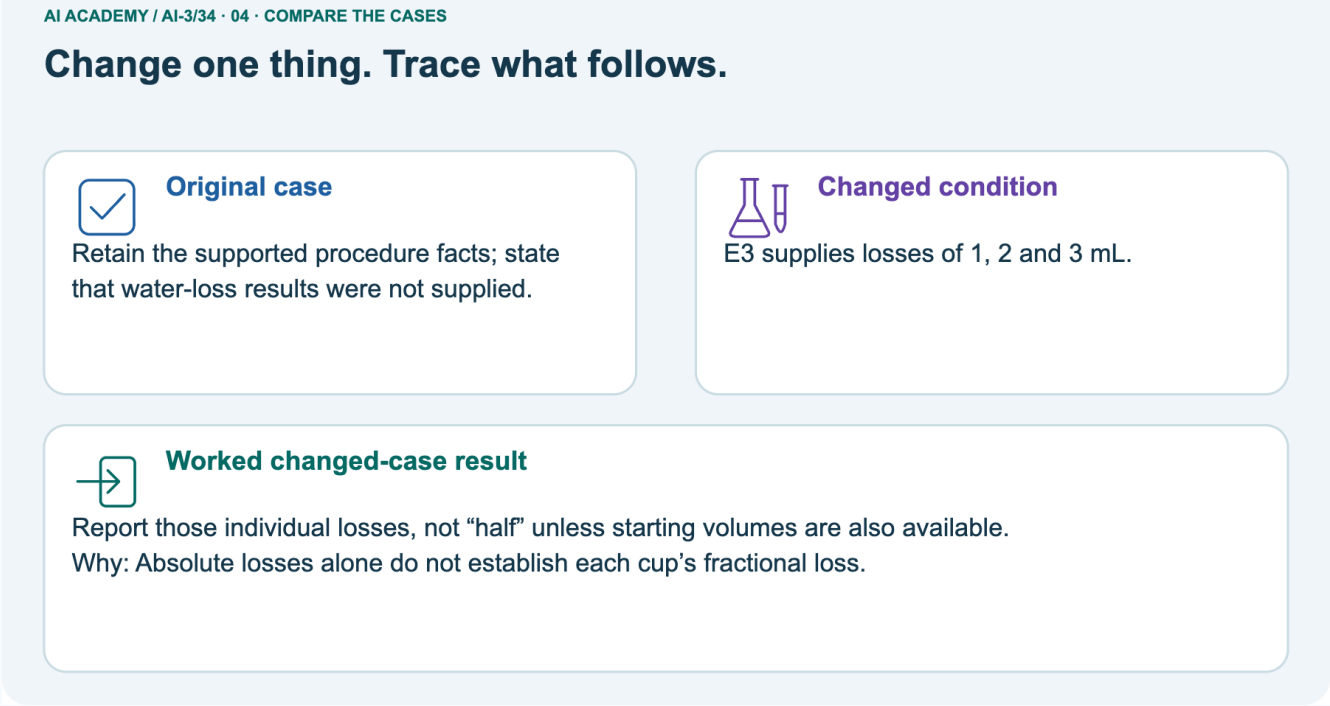
```
def build_report(record):
    # Short fictional classroom records, not arbitrary text or live measurements.
    fields = {'source_id', 'cups', 'days', 'losses_ml', 'initial_ml'}
    if type(record) is not dict or set(record) != fields:
        raise ValueError('exact record fields required')
    if type(record['source_id']) is not str or not record['source_id']:
        raise ValueError('source ID required')
    for key in ['cups', 'days']:
        if type(record[key]) is not int or record[key] < 1:
            raise ValueError('positive integer counts required')
    losses = record['losses_ml']
    initial = record['initial_ml']
    for values, minimum in [(losses, 0), (initial, 1)]:
        if values is not None:
            if type(values) is not list or len(values) != record['cups']:
                raise ValueError('one value per cup required')
            if any(type(x) is not int or x < minimum for x in values):
                raise ValueError('integer mL values outside the toy contract')
    if initial is not None and losses is None:
        raise ValueError('initial values do not supply losses')
    if initial is not None and any(x > y for x, y in zip(losses, initial)):
        raise ValueError('loss cannot exceed initial volume')
    half = None
    if losses is not None and initial is not None:
        half = all(2 * loss == start for loss, start in zip(losses, initial))
    return {'source_id': record['source_id'],
            'procedure': {'cups': record['cups'], 'days': record['days']},
            'losses_ml': losses, 'all_lost_half': half}

E2 = {'source_id': 'E2', 'cups': 3, 'days': 4,
      'losses_ml': None, 'initial_ml': None}
E3 = {'source_id': 'E3', 'cups': 3, 'days': 4,
      'losses_ml': [1, 2, 3], 'initial_ml': None}
print(build_report(E2))
print(build_report(E3))
```

Predict adding E3 losses without starting volumes

- Change one thing: E3 supplies losses of 1, 2 and 3 mL. Predict the result before checking the explanation. What remains the same?

▼ Check after predicting



Follow the labelled picture with the matching explanation. Open the vector diagram

State invented-data and scope limits

- The examples are invented **observation**s; no real classroom investigation or model run is claimed.

- Read the labelled picture: What additional numbers would a half-water claim require? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

▼ Review · Room B/C evidence mismatch

Request room B; retrieved approval names B but register names C. Draft gives capacity 10 from C.

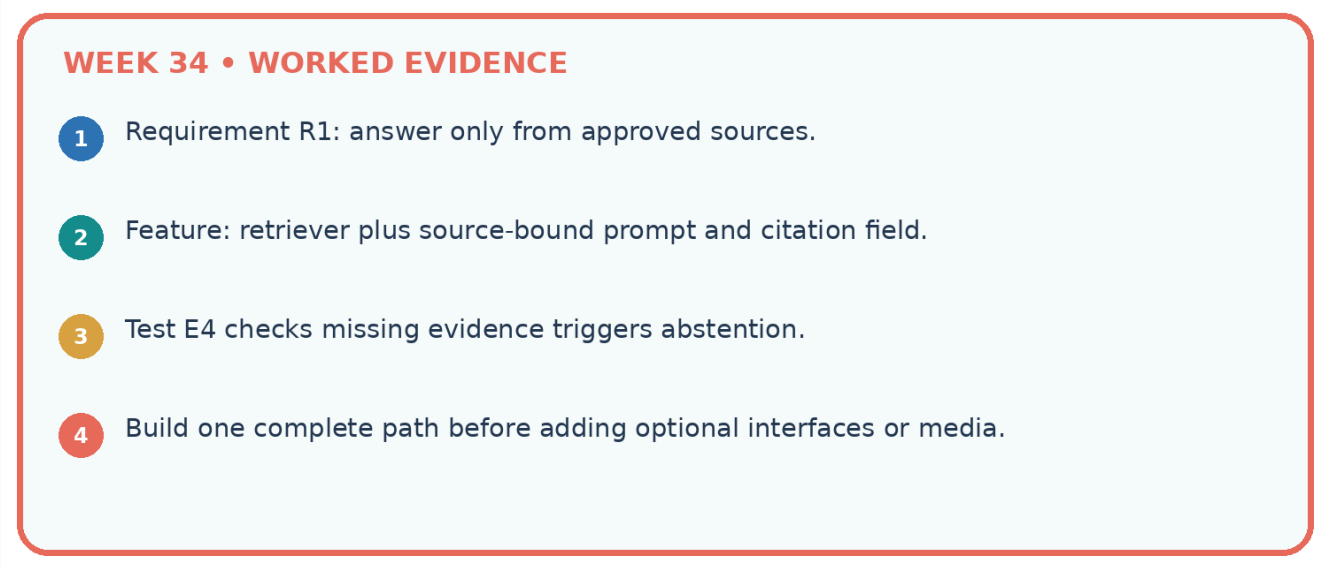
Same-room support fails before or during semantic validation. Return needs_review with null claims; the draft capacity is unsupported for B.

▼ Earlier wording, alternatives and picture transcripts

WORKED **EVIDENCE**

Reason Step by Step

Predict first. Then check each step. A correct answer without visible reasoning is incomplete **evidence**.



1. Requirement R1: answer only from approved sources.

2. Feature: retriever plus source-bound prompt and citation field.

3. Test E4 checks missing **evidence** triggers abstention.

4. Build one complete path before adding optional interfaces or media.

Recalculate, retrace, or restate the reasoning in your own representation:

Write or draw your reasoning and evidence.

CHECK Did the **evidence** answer the exact question? Did the total, path, or source record stay consistent?

A closer explanation

Compare the intermediate **evidence** with the final statement. Explain why each number or decision follows from the stated **input**s.

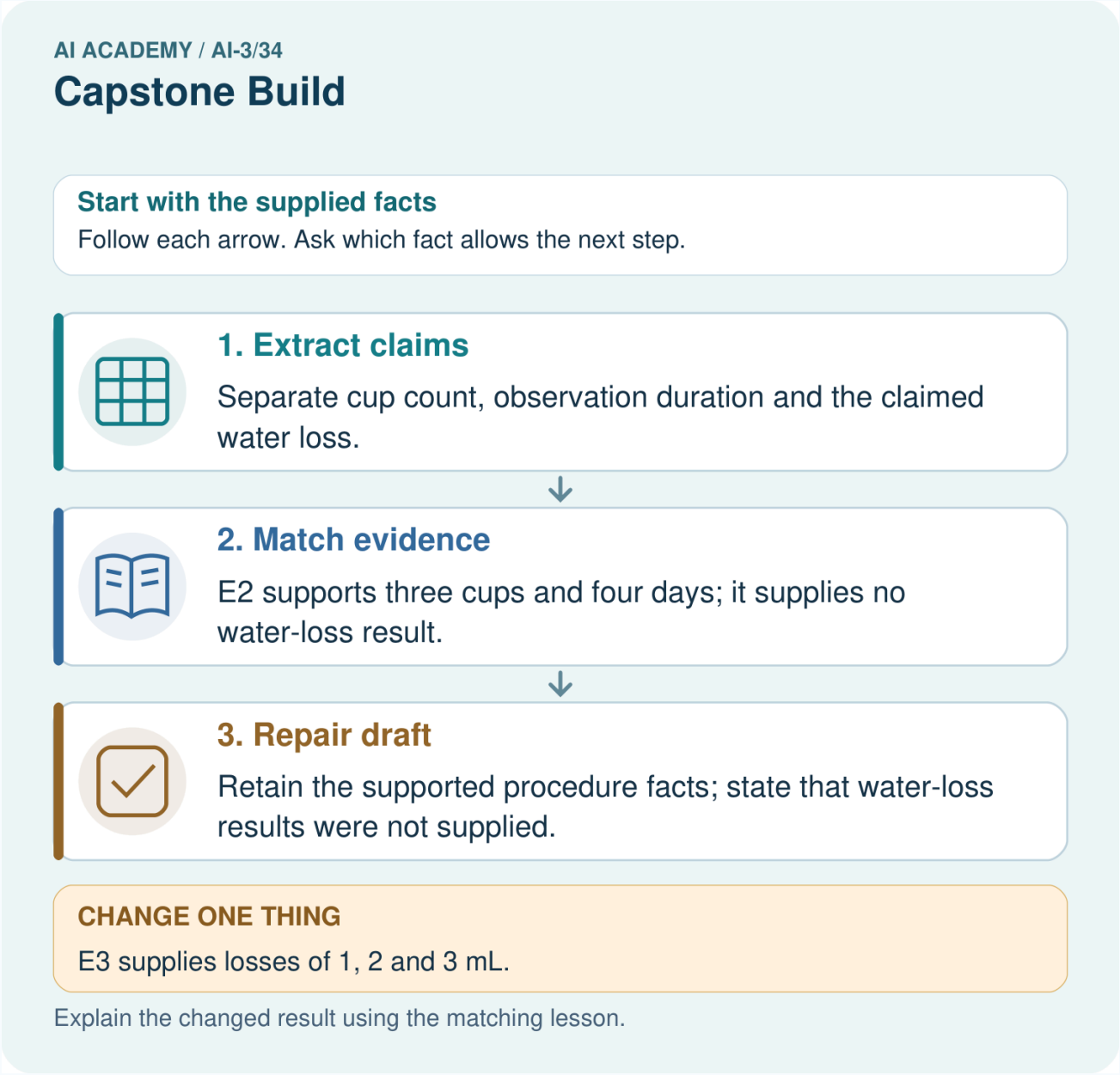
The conclusion belongs to this scenario and procedure; changing the **input**s, evaluation population or authority can change what it establishes. Retain the raw record so a later revision can be compared honestly.

Learn from the worked case

Cover the result before checking it. Say what is given, choose the procedure, and follow one step at a time. Then uncover the result and explain your first mismatch. Ask why a step is allowed, not only what number it produces.

A pictorial worked case: Capstone Build

The facts we will use



Follow the labelled picture with the matching explanation. Open the vector diagram

Why this result follows

What this example does not establish

MAIN LESSON · SECTION 7

Find a result added during formatting

> A clear report should preserve an unknown rather than hide it.

Repair ratio and missing-as-zero shortcuts

Claim to repair: E3 gives losses 1, 2, 3 mL, so every cup lost half its water. A fraction needs the corresponding initial volume. Loss 1 mL could be half of 2 mL but not half of 10 mL. Without initial volumes, report the losses as supplied and the ratio as unresolved. Another shortcut is replacing missing measurements with 0; that fabricates an observed zero. Retain None/null or an explicit unavailable status.

Design a revealing follow-up

> **Exit: propose one revealing change to the worked scenario. State the expected outcome and the evidence that would make you revise your explanation.**

▼ **Earlier wording, alternatives and picture transcripts**

ERROR DETECTIVE

Compare → Diagnose → Repair

Do not patch the answer. Find the earliest unsupported assumption, wrong step, or weak **evidence**.

!

COMPARE • DIAGNOSE

Weak idea: More features create a more complete first build. Strong move: inspect the claim, source, stage, schema, or test evidence; repair one cause; rerun.

Evidence record	Expected	Actual	Likely cause / next test
normal case	works		
boundary case	stated rule		
missing input	safe response		
fresh transfer	generalizes		

Repair one cause. Why is this repair better than changing several things?

Write or draw your reasoning and evidence.

A closer explanation

Complete the supported workbook tasks before attempting W4 independently. Record any hints or help used. After feedback, use the teacher's fresh retry rather than copying the earlier answer.

For a project, select a relevant new case, predict the outcome from the declared rule and preserve the result alongside any revision.

MAIN LESSON · SECTION 8

Practise the earlier room mismatch with less help

> This repeated development case is separate from the fresh seed investigation.

The happy-path limit

A correct happy path does not establish fallback, access control or side-effect safety.

Complete room B/C W2

Practise with less help: Trace the worked case

Use workbook W2 for the connected example “Capstone grounded assistant”. First fill the **input** and rule boxes. Try the middle steps before asking for a hint. After a hint, try the next step yourself.

Plan	Do	Check
What is given? Which rule or procedure applies?	Record each intermediate step.	Does the result follow? What remains unknown?

Audit an all-future-cases claim

- Use the worked case for Capstone grounded assistant. Proposed explanation: “The worked result establishes every future case”. Decide whether this explanation is supported.

  Point to the step or supplied fact that decides; explain your repair if needed.

- Answer on your own before discussion. If you are unsure, say which step you need help with.

Check your reasoning

Explain the pictured missing initial volumes

- What additional numbers would a half-water claim require? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

Explain the picture

> **My explanation using a pictured step and the given facts**
>
> | |
> |---|
> | |
> | |
> | |
> | |

Keep independent work separate

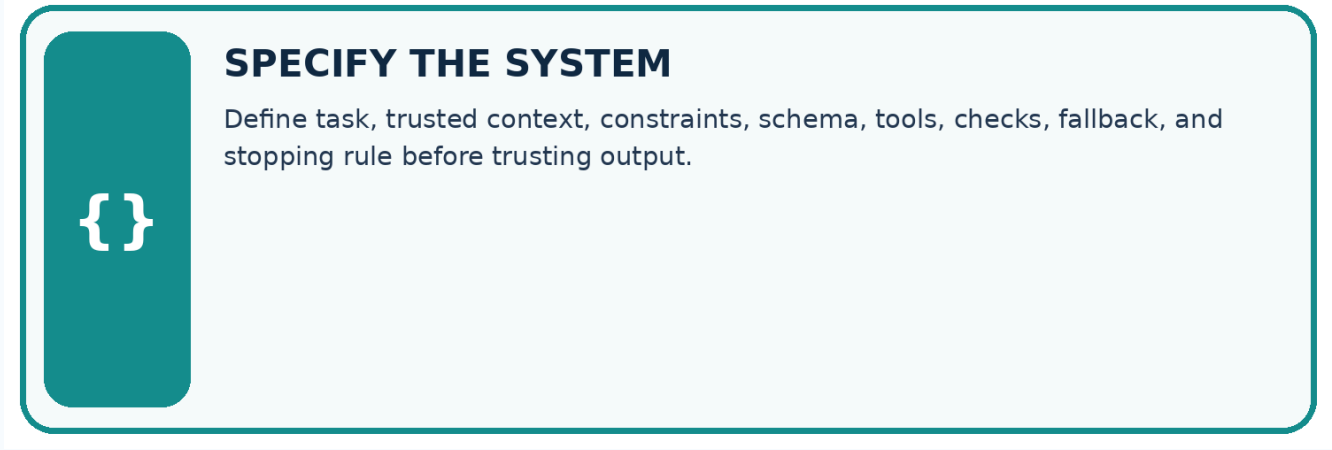
This is supported practice. Keep the separately named independent case for an attempt before feedback.

▼ **Earlier wording, alternatives and picture transcripts**

LOGIC STUDIO

Specify Before You Run

Prompts, schemas, retrieval, tool calls, and code are inspectable parts of one AI product.



SET task, audience, constraints, **output**_schema LOAD approved_context or retrieval_result CALL model or allowed_tool VALIDATE fields, claims, citations, and safety IF **evidence** missing: ask, retrieve, or abstain RETURN useful **output** + source trail + limitations

Trace one **input**. Which line changes the state or chooses a path?

> **Write or draw your reasoning and evidence.**

A closer explanation

A useful check asks what would make the reasoning fail. Distinguish a failure to execute the declared procedure from a weakness in the procedure itself.

The first may require a correction; the second requires an explicitly revised method and new **evidence**. Preserve unsuccessful cases so an improved result cannot hide the original problem.

Guided: restate the worked result, then name the assumption or limitation that prevents a broader conclusion.

Explain your trace to a partner. Your partner points to one step to check. Swap roles on the next case. Each person writes their own explanation.

MAIN LESSON · SECTION 9

Integrate and check five complete report cases

> A smoke check shows a narrow path still works; it is not a broad evaluation result.

Requirement-to-component test cases

Requirement / complete case	Expected build check
R1 · E2 source record with no measurements	Report cups 3/days 4/source E2; losses and half status unavailable.
R2 · E3 adds losses[1,2,3], no initial volumes	Report those losses; half status remains unavailable.
R3 · Proposed test E4 adds initials [2,4,6] to E3	Predict all_lost_half=True under supplied values; this is a new fictional input, not an actual experiment.
R4 · Proposed test E5 losses[0,0,0], initials[2,4,6]	Measured zeros remain zeros; all_lost_half=False.
R5 · Proposed malformed record has only 2 losses for 3 cups	Reject record length mismatch; no accepted report.

Build, predict, trace or execute, and preserve failures

1. Connect the source-card reader, structured-record validation, report builder and final claim check. Link each component to the stated requirement and case. Predict E2/E3, then run or trace the supplied complete program and record which you did. Add E4/E5 as explicit fictional variants; preserve predicted values separately from actual execution. Run a short integration check after each change and retain errors.

   For R5, diagnose the failed input contract instead of padding a missing loss with zero. The program does not read arbitrary model text or authenticate experiments; any natural-language claim still needs evidence review. Keep the independent seed note withheld.

Keep the integration evidence

Test / input · Prediction · Actual · Evidence / calculation · Change or keep?

> **Test / input:** normal
>
> **Prediction:** _____
>
> **Actual:** _____
>
> **Evidence / calculation:** _____
>
> **Change or keep?:** _____

> **Test / input:** boundary
>
> **Prediction:** _____
>
> **Actual:** _____
>
> **Evidence / calculation:** _____
>
> **Change or keep?:** _____

> **Test / input:** missing
>
> **Prediction:** _____
>
> **Actual:** _____
>
> **Evidence / calculation:** _____

> **Change or keep?:** _____

> **Test / input:** fresh transfer
>
> **Prediction:** _____
>
> **Actual:** _____
>
> **Evidence / calculation:** _____
>
> **Change or keep?:** _____

What did the hardest test teach you that the normal test did not?

> **Write or draw your reasoning and evidence.**

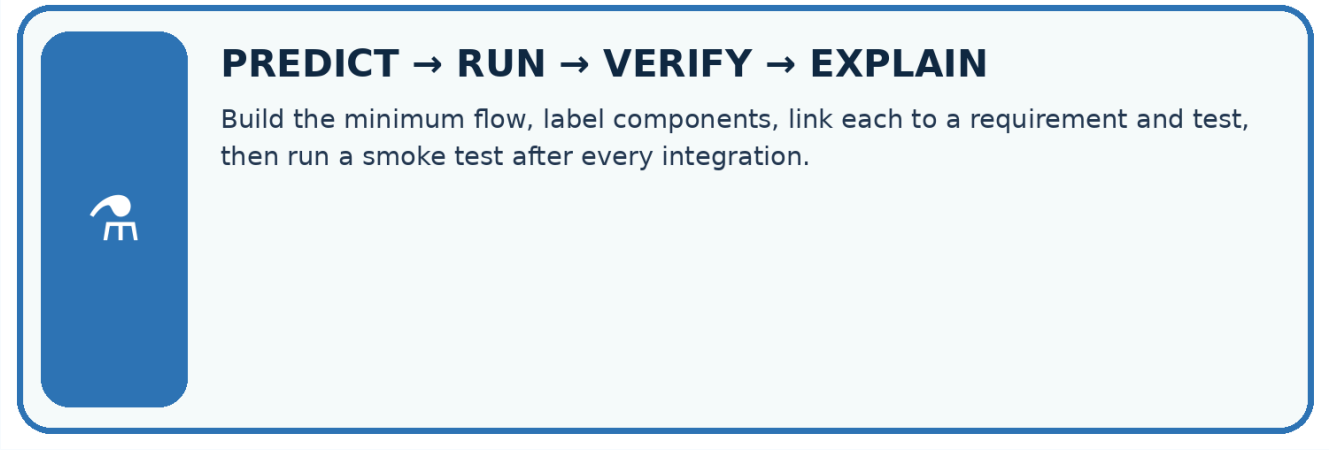
Before trying: what do I expect, and why? While trying: which step could first go wrong? Afterwards: what did I actually observe, and what should change? Keep “not run” if you only traced the procedure.

▼ Earlier wording, alternatives and picture transcripts

EVIDENCE LAB

Predict → Test → Record → Explain

Keep expected and actual results separate. Surprises are useful data when recorded honestly.



Use the complete structured E2/E3 report builder and requirement-linked variants E4/E5/mismatched length. Connect validation, source-preserving output and claim review, recording run/trace status after **integration** changes. Do not turn absent measurements into zero or infer half-water ratios without initial volumes.

Plan → try → check

MAIN LESSON · SECTION 14

Try alone: repair the fresh seed-investigation draft

> Procedure facts do not prove the outcome or every future condition.

Independent source, draft and scope

An experiment note records four fictional seeds observed for six days but supplies no growth results. A generated draft says all four sprouted and the method always works in any soil. Repair it.

Use only the independent experiment's stated procedure facts and missing results. No sprouting observations, soil comparison or later follow-up is supplied. Explain which claims may remain and which require new evidence; do not substitute the cup example's measurements into this seed task.

Your first attempt and help record

My independent explanation, trace and limits

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

Your teacher has the independent answer and scoring criteria. Complete the matching workbook week to keep your practice **evidence** together.

Use feedback afterwards

Attempt the independent case before hints. Record whether you worked alone, used a hint or followed a model. After feedback, fix the first unsupported step and explain why it changed.

A later check uses a changed case, not a copied answer.

▼ **Earlier wording, alternatives and picture transcripts**

Independent case: Changed independent case

Week 34 Independent application

Close the worked explanation. Apply the idea to this different question without copying.

The interface looks complete, but its answer button returns the same saved sentence for every question. What capability is still missing, and how can you expose it?

My answer, with a trace, calculation or supporting **evidence**

> **Write or draw your reasoning and evidence.**

Create and test

Change one relevant detail. Predict the result before testing. Include a boundary or missing-information case when it helps reveal a limit.

My changed **input** and expected result

> **Write or draw your reasoning and evidence.**

Connect to your project

Assemble interface, retrieval, prompt, model/tool, validation, **evidence** display, and fallback.

The **evidence** I will keep

> **Write or draw your reasoning and evidence.**

Transfer practice from the original programme

A prototype cites note A but its answer adds a detail found nowhere in A. What should the final validation stage check and do?

Explain the mechanism, evidence and one limit.

Use feedback to improve

MAIN LESSON · SECTION 11

Keep the project's implementation traceable to its design

> Every changed component needs a matching narrow check and preserved evidence.

This week's contribution

For the Verified Study Buddy, preserve a build version, component map and linked requirements/tests. Save source IDs, parsed facts, missing fields, generated or deterministic draft, validation and displayed answer separately. Keep implementation failures and source gaps distinguishable.

Record a deterministic baseline honestly; adding AI wording later requires a further support check rather than an assumed improvement.

For your Verified Study Buddy, save the **input**, procedure, observed result and limitation of this check.

Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding.

Retain checkpoint and human choice

THIS WEEK'S CHECKPOINT Assemble interface, retrieval, prompt, model/tool, validation, **evidence** display, and fallback.

SYSTEM MAP Learner need → prompt/schema → approved sources/retrieval → model/tool → structured answer → verification → **human decision**

What will you add, change, or verify in the project this week?

Write or draw your reasoning and evidence.

What should remain under **human control**?

Write or draw your reasoning and evidence.

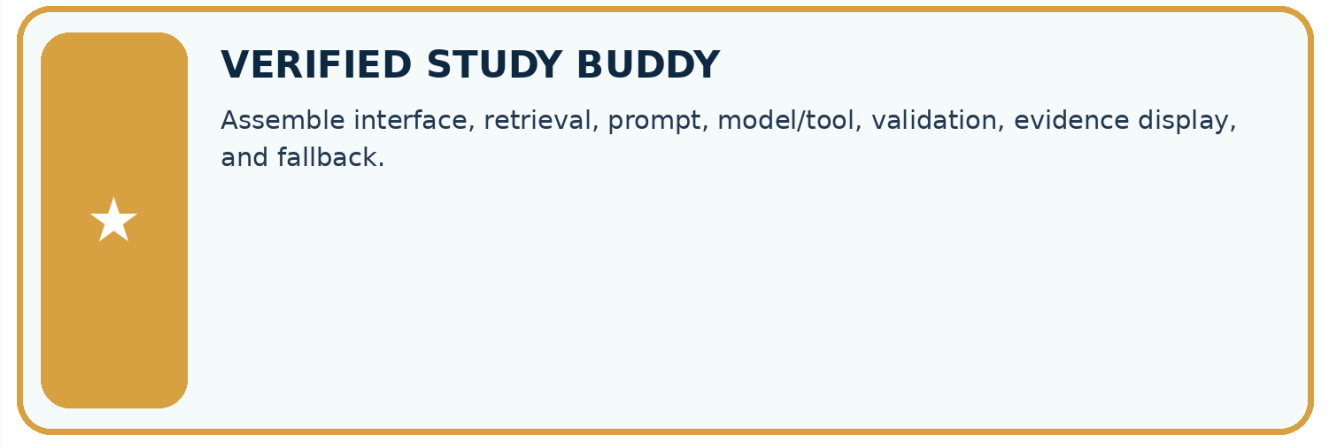
RESPONSIBLE DESIGN Collect only necessary information. Show uncertainty and limits.
Never confuse fluent **output** with verified truth.

▼ Earlier wording, alternatives and picture transcripts

CUMULATIVE PROJECT

Verified Study Buddy

Every week adds one inspectable piece. The system must show how it reached a result.



Evidence for your project

MAIN LESSON · SECTION 16

Finish: state what the build preserved and what remains unknown

> A valid report does not create missing measurements.

Explain without notes

- Close your notes. Explain this goal in your own words: Explain the **mechanism** and terminology of capstone grounded assistant

My deciding step, evidence and one remaining question

Plan a changed-case return

If you are unsure, return to the worked picture and explain one arrow. If you are ready, change an **input** or assumption and justify your **prediction**.

Save your workbook **evidence**. At the next review, try a different case before opening your old answer.

Reference bank and extra practice

Choose the reference that helps your next step.

▼ **Your quick word guide**

REFERENCE · SECTION 5

Your quick word guide

> Use the words to explain a particular step, record or limit.

Core vocabulary

Word	My own example	Not this / boundary
traceability	The ability to connect a result back to its inputs, sources, steps and version.	
integration	Connecting components so that they work together through agreed interfaces.	
version control	A method for recording and comparing changes to identified versions of files.	
minimum complete product	The smallest working version that fulfils its essential user task, including failure handling.	

Terms for the connected explanation

Word	Meaning
Pipeline stage	One operation with defined inputs and failure handling

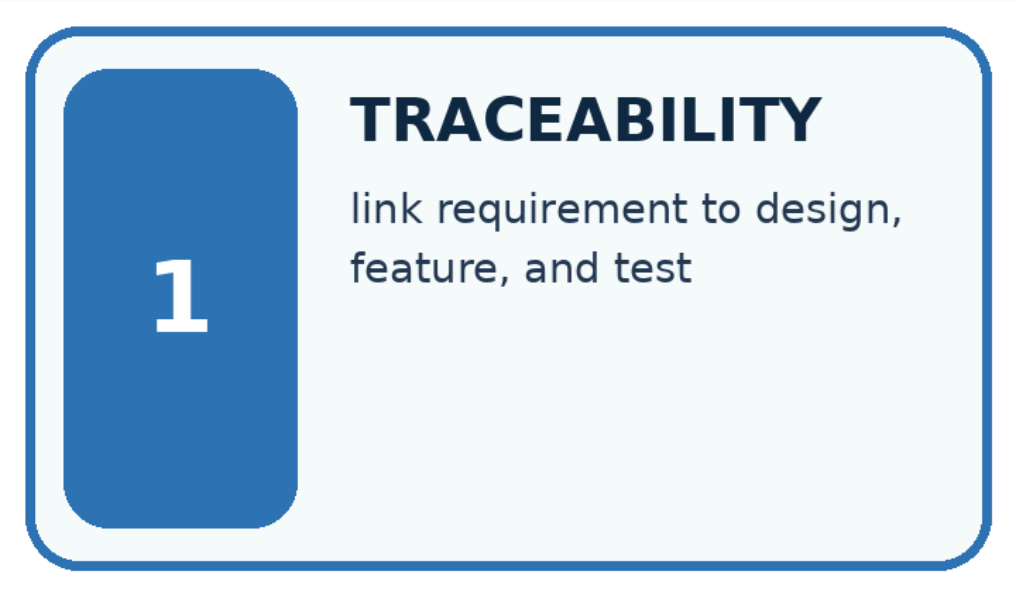
Word	Meaning
Answerability gate	A check for sufficient evidence before answering
Semantic validator	A check that claims match task evidence

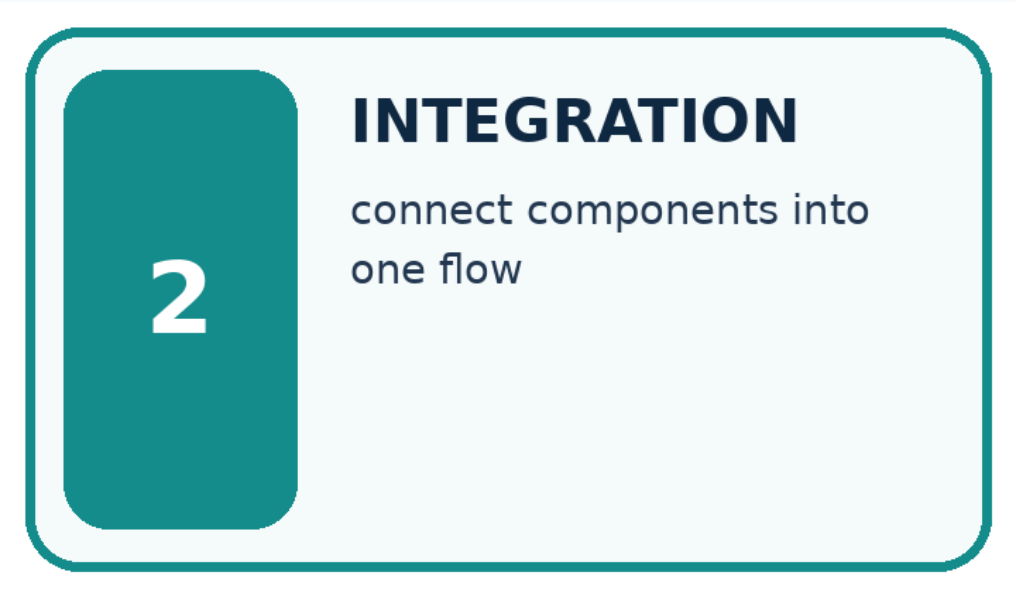
▼ Earlier wording, alternatives and picture transcripts

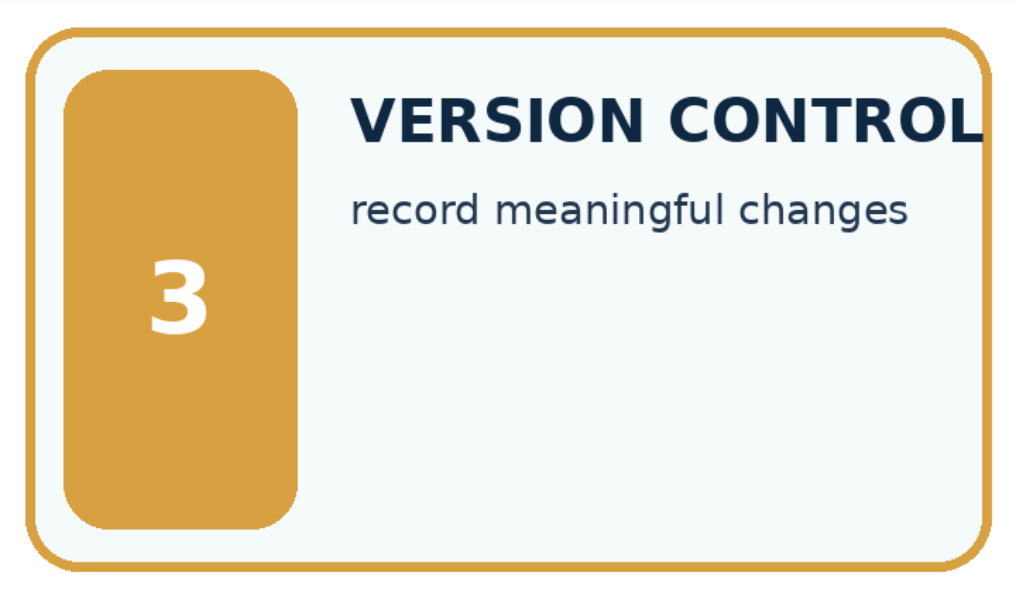
VOCABULARY LAB

Words You Can Use to Reason

A definition matters when you can recognize the idea, use it, and contrast it with a near-miss.




2
INTEGRATION
connect components into one flow


3
VERSION CONTROL
record meaningful changes



Words used in the closer explanation

▼ **Choose a stronger thinking challenge**

REFERENCE · SECTION 10

Choose a stronger thinking challenge

> Choose the next useful challenge rather than treating the whole ladder as compulsory.

Pick your next step

- 1 • NAME: Define the concept without repeating the textbook.
- 2 • TRACE: Show the information, state, branch, calculation, or **evidence** path.
- 3 • APPLY: Solve a different example independently.
- 4 • CHALLENGE: Find a boundary case, counterexample, or unfair outcome.

- 5 • EXPLAIN LIMITS: Say when the result should not be trusted or used alone.

- My strongest **evidence** of independent understanding:

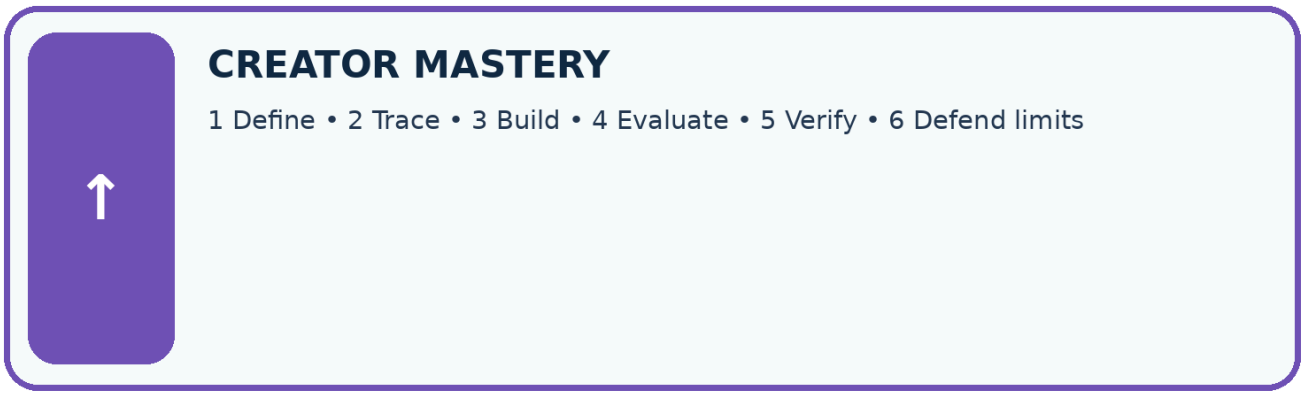
Write or draw your reasoning and evidence.

▼ Earlier wording, alternatives and picture transcripts

MASTERY

From Knowing the Word to Owning the Idea

Move upward only when you can produce **evidence** without copying the worked example.



▼ Reference: check your explanation

Reference: check your explanation

> Use after practice. Repair the new seed-investigation draft independently is the selected independent case.

Check the evidence and its limit

- 1. My fresh example, trace, calculation, or design:
- 2. My **evidence** that it works (include a boundary or missing-data case):
- 3. One limitation, fairness concern, or situation needing human review:

> **Write or draw your reasoning and evidence.**

> **Write or draw your reasoning and evidence.**

Self-check and record

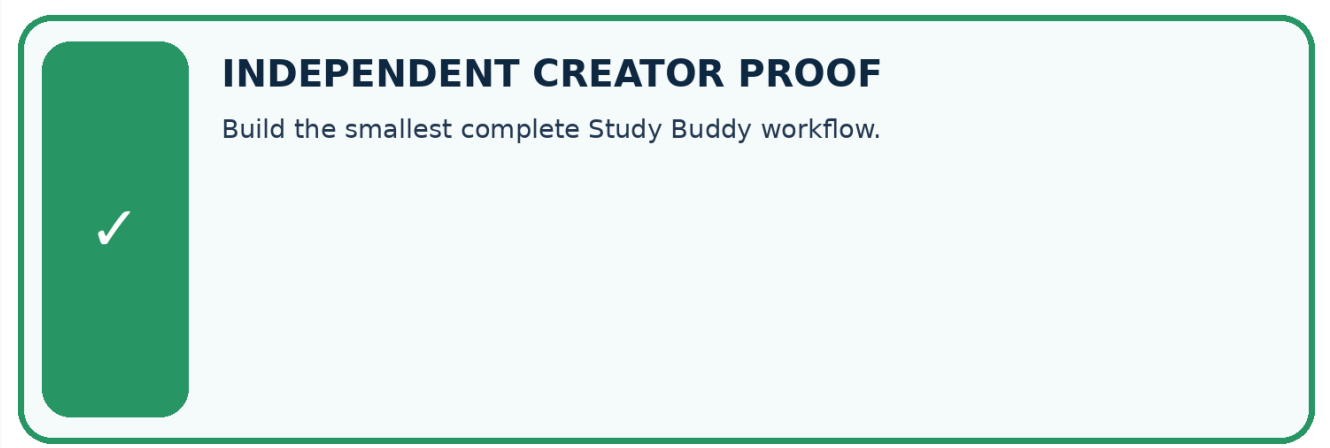
SELF-CHECK ☐ I explained the **mechanism** ☐ I used **evidence** ☐ I tested a fresh case ☐ I named a limitation ☐ I can answer 'why?'

▼ **Earlier wording, alternatives and picture transcripts**

INDEPENDENT MASTERY PROOF

Checklist for explaining this week's idea

Use after practice. Repair the new seed-investigation draft independently is the selected independent case.



EXPLANATION CHECK Require supported procedure facts, missing-result status and rejection of unsupported universal conclusions.

Write or draw your reasoning and evidence.

▼ **Extra practice: Validate before displaying the draft**

REFERENCE · SECTION 13

Extra practice: Validate before displaying the draft

> Try the question before opening its worked explanation.

Read the supplied case and question

A pipeline retrieves passages, drafts and displays without checking citations.

What step is missing?

Your explanation

- Which detail supports the answer, and why does it matter?

Write or draw your reasoning and evidence.

▼ Check after trying

Validate every factual claim against its citation, plus required fields and boundaries. Revise or stop unsupported answers; retain diagnostic stages.

Evidence or stage	Value or result
Retrieve	Relevant approved passages
Draft	Compose the answer
Validate	Check claims and citations
Display or stop	Show supported content or explain the limit

▼ Earlier wording, alternatives and picture transcripts

Week 34 Worked case

Use the supplied details to practise the weekly concept before attempting the independent question. This case extends the illustrated lesson.

The situation

The question

Worked explanation

Explain it in your own words

Trace the supplied case visually

Validate before displaying the draft

A pipeline retrieves passages, drafts and displays without checking citations.

What step is missing?

Validate every factual claim against its citation, plus required fields and boundaries. Revise or stop unsupported answers; retain diagnostic stages.

▼ Reference: original lesson checklist

REFERENCE · SECTION 15

Reference: original lesson checklist

> Use an additional checklist when it helps inspect your explanation.

Targets and evidence

By the end, I can	How I will show it
Explain the mechanism and terminology of capstone grounded assistant	Explain the deciding step, show a trace or test, and state one limit.
Apply the stated procedure to a new case with a traceable result	Explain the deciding step, show a trace or test, and state one limit.
Identify an assumption or limitation and design a revealing follow-up	Explain the deciding step, show a trace or test, and state one limit.

Keep your route and first prediction

Remember → watch and explain → try with help → investigate → try independently → keep project **evidence**. Your teacher may spread this lesson across several classes.

Keep your first **prediction**. Compare it with what the **evidence** shows and explain any change.

▼ Earlier wording, alternatives and picture transcripts

Start with this question: Build a claim checker for a draft summary of a fictional class investigation.

▼ Extra practice: pictures, repair and feedback

REFERENCE · SECTION 17

Extra practice: pictures, repair and feedback

> Attempt each question before opening its explanation. Use one browser or paper copy.

First supported question

- Build a claim checker for a draft summary of a fictional class investigation.
- Which labelled step matches this detail? Retain the supported procedure facts; state that water-loss results were not supplied.
- 1. Extract claims
- 2. Match **evidence**
- 3. Repair draft
- Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.
- After your attempt, compare with the supported explanation: Repair draft Retain the supported procedure facts; state that water-loss results were not supplied. Read the picture in order.

  Retain the supported procedure facts; state that water-loss results were not supplied. The arrows show which recorded **input** or intermediate result each decision uses.

  Explain the deciding step aloud before checking the changed case; pointing to the final answer alone does not show how it follows.

Practice 1: explain the deciding evidence

My selected conclusion, deciding fact and explanation

▼ Check the first explanation

Practice 2: explain the deciding evidence

Changed-case question

- Changed-case practice: E3 supplies losses of 1, 2 and 3 mL. Which conclusion is supported by the supplied facts?
- 1. We can decide the result without examining the changed **input** or the procedure.
- 2. Report those individual losses, not “half” unless starting volumes are also available.
- 3. This one example guarantees correct results for every future **input**.
- Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.
- After your attempt, compare with the supported explanation: Report those individual losses, not “half” unless starting volumes are also available. Absolute losses alone do not establish each cup’s fractional loss.

My selected conclusion, deciding fact and explanation

▼ Check the changed case

Repair a missing building block

Repair and save your work

Review Capstone Design (ai-3/33). Goal: Explain the **mechanism** and terminology of capstone requirements and corpus. Short practice: What should the project plan record before the new flower test?

Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.. Return to this week and explain what you changed.

Review Safety Boundaries (ai-3/31). Goal: Explain the **mechanism** and terminology of untrusted **input** boundaries. Short practice: If AB7 passes the pattern, may it access teacher-only material?

Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.. Return to this week and explain what you changed.

Review Privacy by Design (ai-3/28). Goal: Explain the **mechanism** and terminology of data retention and deletion. Short practice: Which IDs and locations must appear in the deletion **evidence**?

Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.. Return to this week and explain what you changed.

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

Drafts stay in this browser when storage is available; clear drafts on a shared device.

▼ Earlier wording, alternatives and picture transcripts

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

Keep your completed work

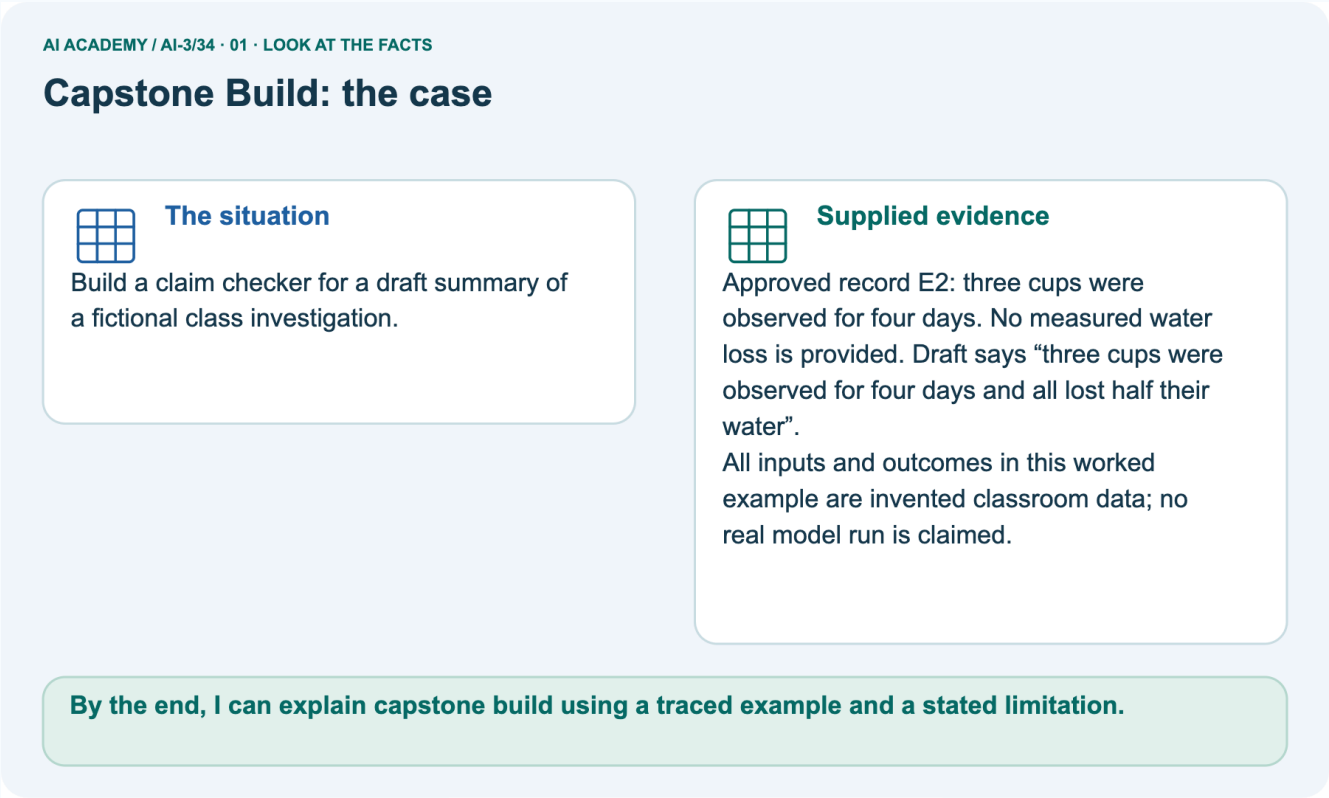
▼ Reference: the six-picture journey

REFERENCE · SECTION 18

Reference: the six-picture journey

> Revisit the picture that serves the current question; key diagrams already sit beside the main explanation.

01. Capstone Build: the case



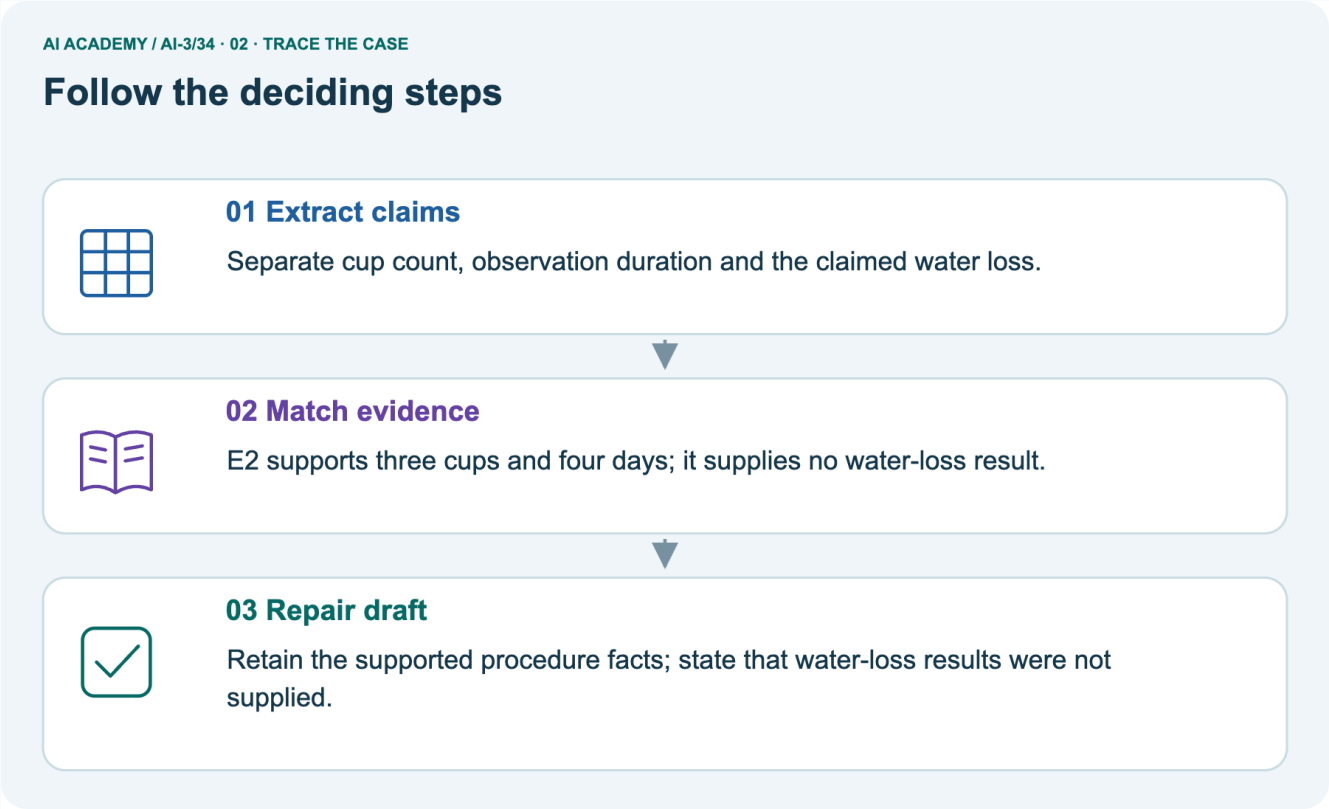
Follow the labelled picture with the matching explanation. Open the vector diagram

Build a claim checker for a draft summary of a fictional class investigation. Supplied facts: Approved record E2: three cups were observed for four days. No measured water loss is provided.

Draft says “three cups were observed for four days and all lost half their water”. All **input**s and outcomes in this worked example are invented classroom data; no real model run is claimed.

Point to the given fact that matters most. Explain why it matters.

02. Follow the deciding steps

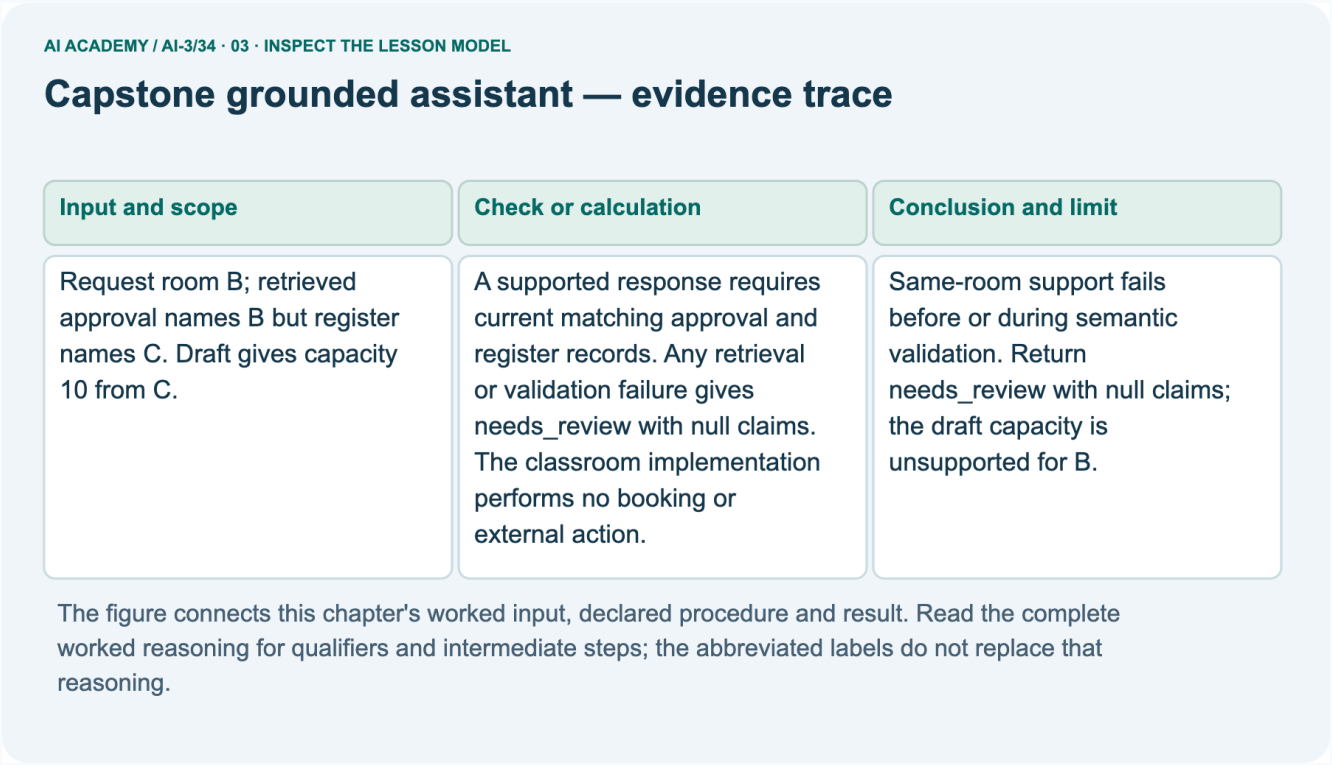


Follow the labelled picture with the matching explanation. Open the vector diagram

Extract claims: Separate cup count, **observation** duration and the claimed water loss. → Match **evidence**: E2 supports three cups and four days; it supplies no water-loss result. → Repair draft: Retain the supported procedure facts; state that water-loss results were not supplied.

Point to each numbered stage and explain which recorded **input** it uses.

03. Capstone grounded assistant — evidence trace



AI ACADEMY / AI-3/34 · 03 · INSPECT THE LESSON MODEL

Capstone grounded assistant — evidence trace

Input and scope	Check or calculation	Conclusion and limit
Request room B; retrieved approval names B but register names C. Draft gives capacity 10 from C.	A supported response requires current matching approval and register records. Any retrieval or validation failure gives needs_review with null claims. The classroom implementation performs no booking or external action.	Same-room support fails before or during semantic validation. Return needs_review with null claims; the draft capacity is unsupported for B.

The figure connects this chapter's worked input, declared procedure and result. Read the complete worked reasoning for qualifiers and intermediate steps; the abbreviated labels do not replace that reasoning.

Follow the labelled picture with the matching explanation. Open the vector diagram

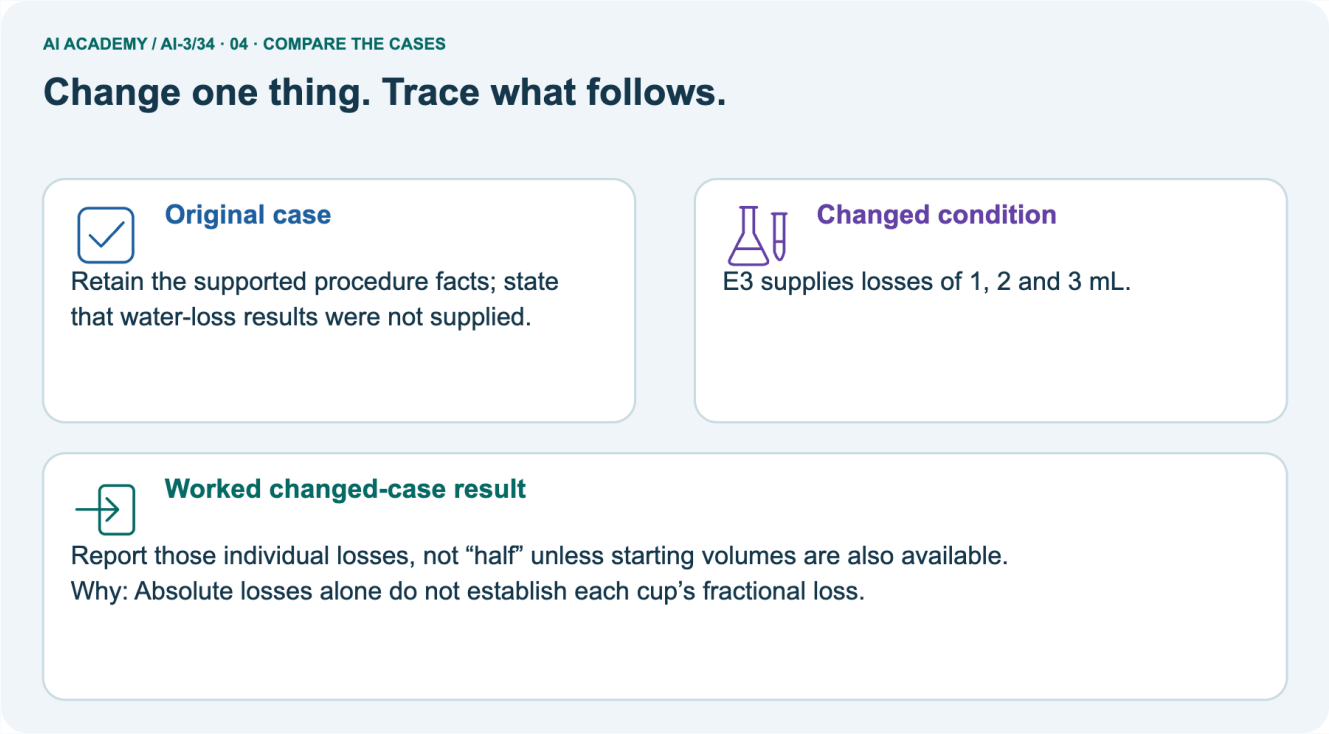
Capstone grounded assistant — **evidence** trace. Request room B; retrieved approval names B but register names C. Draft gives capacity 10 from C. | A supported response requires current matching approval and register records.

Any retrieval or validation failure gives needs_review with null claims. The classroom implementation performs no booking or external action. | Same-room support fails before or during semantic validation.

Return needs_review with null claims; the draft capacity is unsupported for B.

Read the labels and values. Explain the relationship the model shows.

04. Change one thing. Trace what follows.



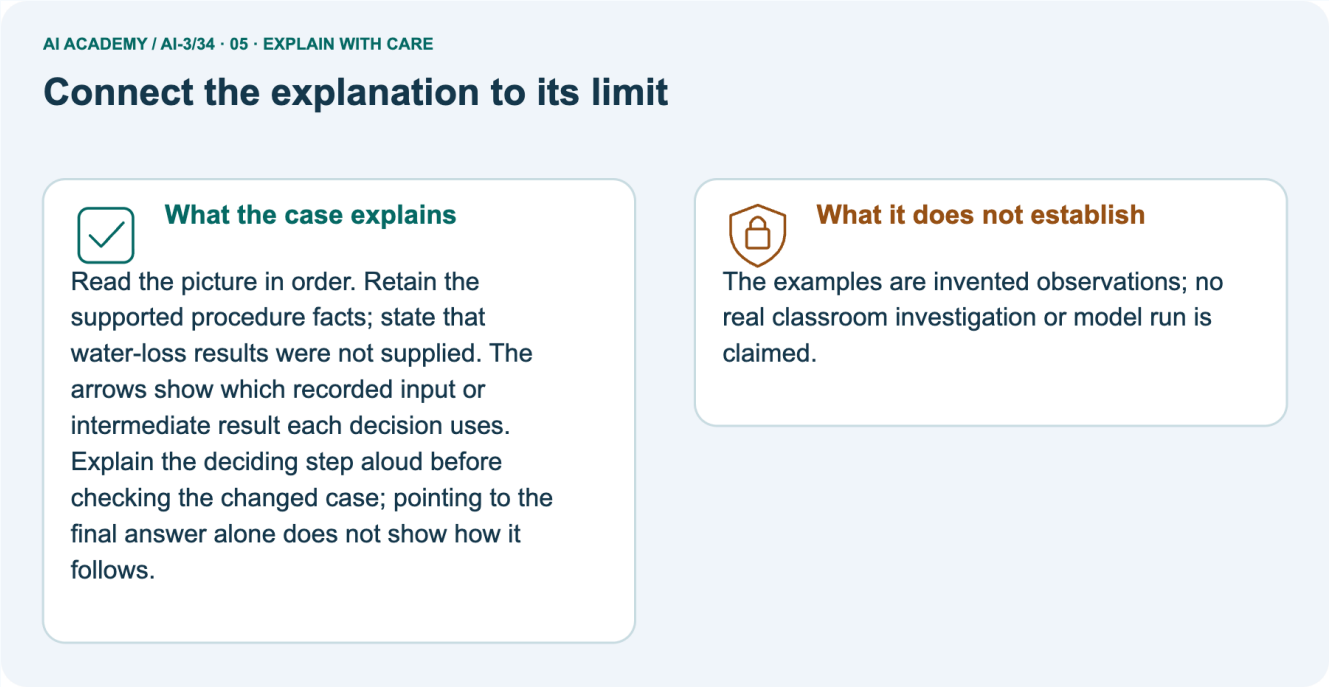
Follow the labelled picture with the matching explanation. Open the vector diagram

Original case: Retain the supported procedure facts; state that water-loss results were not supplied. Changed condition: E3 supplies losses of 1, 2 and 3 mL.

Expected result: Report those individual losses, not “half” unless starting volumes are also available. Reason: Absolute losses alone do not establish each cup’s fractional loss.

Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

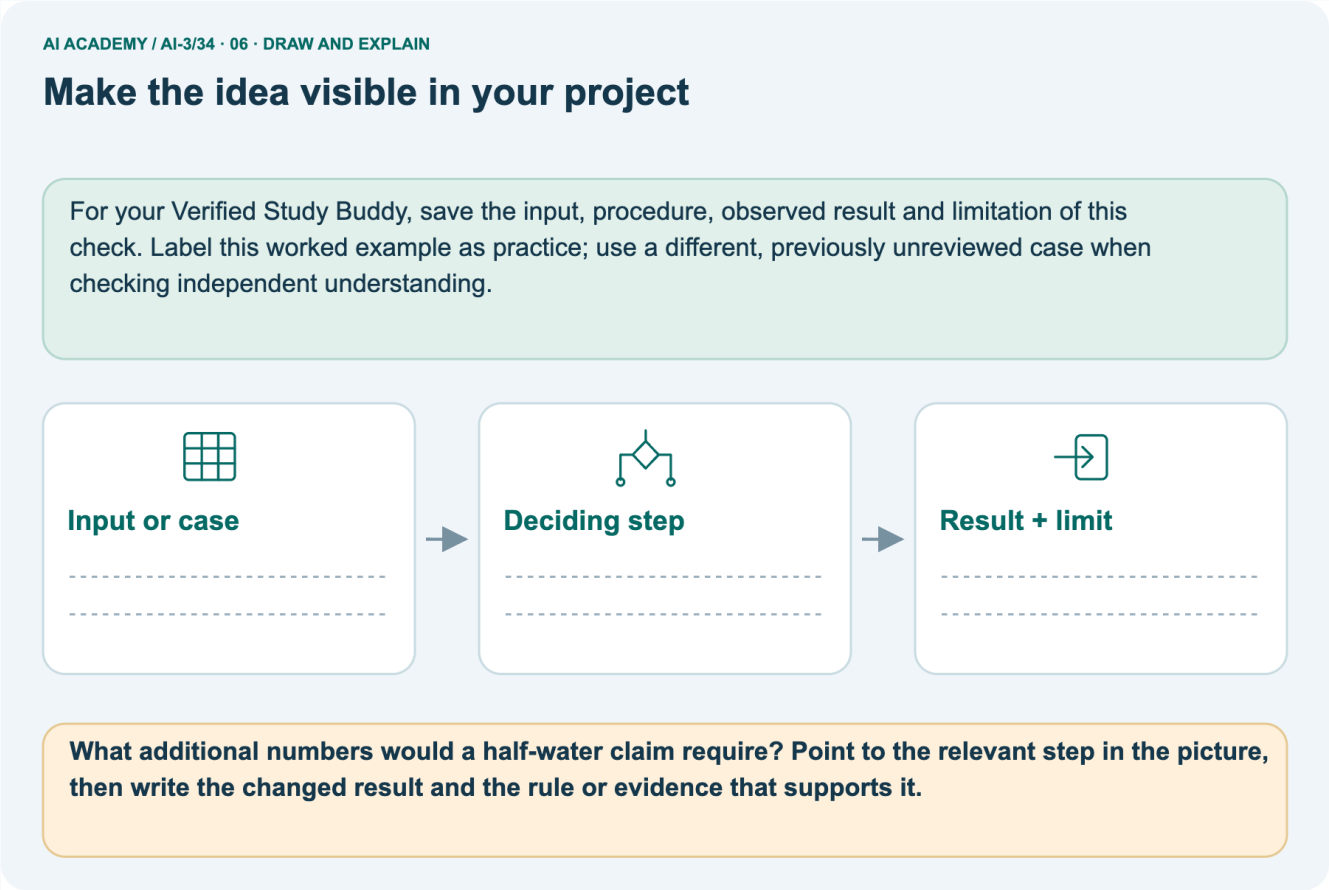
Read the picture in order. Retain the supported procedure facts; state that water-loss results were not supplied. The arrows show which recorded **input** or intermediate result each decision uses.

Explain the deciding step aloud before checking the changed case; pointing to the final answer alone does not show how it follows.

Limit: The examples are invented **observation**s; no real classroom investigation or model run is claimed.

Explain the case, then point to a limit on the claim.

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

For your Verified Study Buddy, save the **input**, procedure, observed result and limitation of this check.

Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding. Draw your **input**, deciding step, result and limit.

Supported question: What additional numbers would a half-water claim require? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

Draw your project trace and justify the deciding step using **evidence**.

▼ Earlier wording, alternatives and picture transcripts

01. Capstone Build: the case

02. Follow the deciding steps

03. Capstone grounded assistant — evidence trace

04. Change one thing. Trace what follows.

05. Connect the explanation to its limit

06. Make the idea visible in your project

▼ Optional interactive practice and completed work

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.